

Quantum Information and Computing

From qubits to quantum key distribution and quantum algorithms — Lecture Notes

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About these notes

Scope

These are lecture notes for *Quantum Information and Computing* (third-semester course, Master in Physics of Data, University of Padova; lectures by Prof. Giuseppe Vallone). The course has three modules:

1. **Fundamentals of quantum mechanics** — Hilbert spaces, pure and mixed states, measurements and POVMs, unitary evolution, composite systems, entanglement, Bell inequalities;
2. **Quantum information** — classical and quantum Shannon theory, no-cloning, quantum key distribution, teleportation and entanglement swapping;
3. **Quantum computation** — classical and quantum circuits, universal gates, the quantum Fourier transform, phase estimation, Shor's and Grover's algorithms, error correction and physical implementations.

Reading the notes

- **Part I (Chapters 1–8):** the postulates of quantum mechanics rewritten for finite-dimensional systems, from a single qubit to open systems. Chapters 5–7 are devoted to entanglement: definition, Bell nonlocality, detection criteria.
- **Part II (Chapters 9–11):** information-theoretic consequences — von Neumann entropy and the Holevo bound, quantum cryptography (BB84 and its descendants, key rates, real implementations, quantum random number generators), teleportation.
- **Part III (Chapters 12–15):** the circuit model, the algorithms that motivate quantum computers, and how to protect and build them.

Every chapter uses the styled callout environments for definitions, theorems, worked examples and remarks, and ends with exercises (solutions collapsible in the HTML version) and a *Sources* section. Diagrams are kept deliberately minimal — the Bloch sphere, a few circuits, protocol sketches — and drawn only where they carry the argument.

Notation

- $|\psi\rangle, \langle\psi|$: kets and bras; $\langle\phi|\psi\rangle$ the scalar product; $\mathcal{H}$ a finite-dimensional Hilbert space of dimension d .
- $\{|0\rangle, |1\rangle\}$: computational basis; $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$; $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$.
- $\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z$ (also X, Y, Z): Pauli matrices; $\vec{r} \cdot \hat{\vec{\sigma}}$ a qubit observable.
- $\hat{\rho}$: density operator; Tr, Tr_B : trace and partial trace.
- $\hat{U}$: unitary operator; $\hat{H}$: Hamiltonian; H (gate): Hadamard.
- $\{\hat{E}_k\}$: Kraus operators; $\{\hat{M}_k\}$: measurement operators; $\{\hat{F}_k\}$: POVM elements.
- $H(X)$: Shannon entropy; $S(\hat{\rho})$: von Neumann entropy; χ : Holevo quantity; $h_2(Q)$: binary entropy.
- $|\Phi^\pm\rangle, |\Psi^\pm\rangle$: Bell states.

Sources

These notes reorganise the course material of Prof. Giuseppe Vallone (University of Padova, A.Y. 2025–26): the lecture summaries (lectures 2–21), the slide decks on quantum key distribution (`Vallone_QKD_lecture1`, `lecture13`, `implementation_lecture14`), quantum computing (`QComputing_25.26`), the “quantum cakes” (`MagicCake_Vallone-expanded`), quantum information research at DEI (`lecture24`), and the homework and exercise sheets. Explanations and examples are drawn from the course textbooks:

- Michael A. Nielsen and Isaac L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed., Cambridge University Press, 2010.
- Giuliano Benenti, Giulio Casati and Giuliano Strini, *Principles of Quantum Computation and Information*, Vol. I, World Scientific, 2004.

with occasional reference to John Preskill’s *Lecture Notes on Quantum Computation* (Caltech, Physics 219). Every chapter closes with its own *Sources* section.

Part I.

Part I: Quantum mechanics for quantum
information

1. Qubits and measurement

Quantum information starts from the simplest quantum system: a two-level system, the **qubit**. This chapter introduces the Dirac notation and the Hilbert-space structure, defines the qubit through the superposition principle, states the Born rule for measurement and the collapse of the state, explains why global phases are unobservable, and pictures every qubit state as a point on the Bloch sphere. The Stern–Gerlach experiment illustrates all of it.

1.1. Kets, scalar products and Hilbert spaces

A quantum state is a vector in a complex vector space, written as a **ket** $|\psi\rangle$. In a d -dimensional space with orthonormal basis $\{|0\rangle, \dots, |d-1\rangle\}$,

$$|\psi\rangle = \sum_{k=0}^{d-1} \psi_k |k\rangle, \quad \psi_k \in \mathbb{C},$$

represented by the column vector $(\psi_0, \dots, \psi_{d-1})^T$.

Definition 1.1 (Scalar product). A scalar (inner) product assigns to each pair of vectors a complex number $\langle\phi|\psi\rangle$ such that:

1. $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$ (conjugate symmetry);
2. it is linear in the second argument: $\langle\phi|a\psi_1 + b\psi_2\rangle = a\langle\phi|\psi_1\rangle + b\langle\phi|\psi_2\rangle$;
3. $\langle\psi|\psi\rangle \geq 0$, with equality iff $|\psi\rangle = 0$.

In an orthonormal basis, $\langle\phi|\psi\rangle = \sum_k \phi_k^* \psi_k$. The norm is $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$.

A (finite-dimensional) **Hilbert space** $\mathcal{H}$ is a complex vector space with a scalar product. (In infinite dimensions completeness must be added; we will rarely need it.)

Definition 1.2 (Bras and Hermitian conjugation). The **bra** $\langle\phi|$ is the linear functional $|\psi\rangle \mapsto \langle\phi|\psi\rangle$. It is the Hermitian conjugate of the ket: as a row vector, $\langle\phi| = (|\phi\rangle)^\dagger = (\phi_0^*, \dots, \phi_{d-1}^*)$. Conjugation is antilinear: $(a|\phi\rangle)^\dagger = a^* \langle\phi|$.

The scalar product is then literally “bra times ket”: a row vector times a column vector.

1.2. The qubit

Definition 1.3 (Qubit). A **qubit** is a quantum system whose states live in a two-dimensional Hilbert space spanned by the **computational basis** $\{|0\rangle, |1\rangle\}$. Its general state is a superposition

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

1. Qubits and measurement

The **superposition principle** says any normalised linear combination of physical states is a physical state. This is what distinguishes a qubit from a classical bit, which is either 0 or 1. Physical realisations include the polarisation of a photon ($|H\rangle, |V\rangle$), the spin of an electron or nucleus ($|\uparrow\rangle, |\downarrow\rangle$), two levels of an atom or ion, two charge or flux states of a superconducting circuit, or two time bins of a photon.

Normalisation $\langle\psi|\psi\rangle = 1$ is the physicality condition, because the coefficients have a probabilistic meaning.

1.3. Measurement and the Born rule

Definition 1.4 (Measurement in an orthonormal basis). Measuring a qubit in state $|\psi\rangle$ in the orthonormal basis $\{|e_0\rangle, |e_1\rangle\}$ gives outcome k with probability

$$p_k = |\langle e_k|\psi\rangle|^2 \quad (\text{Born rule}).$$

Immediately after the measurement the state is $|e_k\rangle$: the state **collapses** onto the basis vector corresponding to the observed outcome.

In the computational basis, $p_0 = |\alpha|^2$ and $p_1 = |\beta|^2$; normalisation guarantees $p_0 + p_1 = 1$. Two features are essential:

- the outcome is **intrinsically random** — not because of ignorance of hidden details, but as a law of nature (this is the resource exploited by quantum random number generators, Chapter 10);
- measurement **disturbs** the state: repeating the same measurement gives the same outcome, but the original superposition is lost.

The same rule works in d dimensions: for $|\psi\rangle = \sum_k \psi_k |k\rangle$ and a measurement in basis $\{|e_j\rangle\}$, $p_j = |\langle e_j|\psi\rangle|^2$ and the post-measurement state is $|e_j\rangle$.

Example 1.1 (Measuring in the diagonal basis)

Let $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. The state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ measured in $\{|+\rangle, |-\rangle\}$ gives

$$p_{\pm} = |\langle \pm|\psi\rangle|^2 = \frac{1}{2}|\alpha \pm \beta|^2.$$

For $|\psi\rangle = |0\rangle$ both outcomes have probability $\frac{1}{2}$: a state perfectly definite in one basis is maximally random in the other. The bases $\{|0\rangle, |1\rangle\}$ and $\{|+\rangle, |-\rangle\}$ are **mutually unbiased**.

1.4. Global phases

For any real γ , the vectors $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ give identical probabilities for every measurement, $|\langle e|e^{i\gamma}\psi\rangle|^2 = |\langle e|\psi\rangle|^2$. They represent the **same physical state**: physical states are rays in $\mathcal{H}$, not vectors. A *relative* phase, on the other hand, is physical: $|+\rangle$ and $|-\rangle$ differ only by the relative sign of $|1\rangle$, yet they are orthogonal and perfectly distinguishable by a measurement in the $\{|\pm\rangle\}$ basis.

Example 1.2 (Same state or not?)

$|1\rangle$ and $i|1\rangle$ are the same state ($i = e^{i\pi/2}$). $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ and $\frac{1}{\sqrt{2}}(|1\rangle - |0\rangle)$ are the same (factor -1). $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ and $\frac{1}{\sqrt{2}}(i|0\rangle - |1\rangle) = \frac{i}{\sqrt{2}}(|0\rangle + i|1\rangle)$ are the same. But $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(-|0\rangle + i|1\rangle)$ are different: in the $\{|\pm\rangle\}$ basis the first gives $p_+ = 1$, the second $p_+ = \frac{1}{2}$.

1.5. The Bloch sphere

Using normalisation and the freedom of the global phase, every qubit state can be written as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad \theta \in [0, \pi], \varphi \in [0, 2\pi).$$

The two angles identify a point on the unit sphere, the **Bloch sphere**, with Cartesian coordinates

$$\vec{r} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta).$$

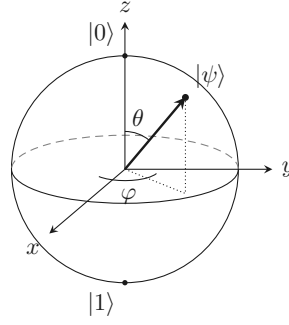


Figure: the Bloch sphere. Poles: $|0\rangle$ and $|1\rangle$; on the equator, $|\pm\rangle$ along $\pm x$ and $|\pm i\rangle$ along $\pm y$.

Key facts:

- the north and south poles are $|0\rangle$ and $|1\rangle$; the equator contains the equal superpositions, with $|\pm\rangle$ on the $\pm x$ axis and $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$ on the $\pm y$ axis;
- **orthogonal states are antipodal**: the overlap of two pure states with Bloch vectors $\vec{r}, \vec{s}$ is $|\langle\psi|\phi\rangle|^2 = \frac{1}{2}(1 + \vec{r} \cdot \vec{s})$, which vanishes iff $\vec{s} = -\vec{r}$ (proved in Chapter 3);
- the angle $\theta/2$ in Hilbert space becomes θ on the sphere: orthogonal vectors (angle $\pi/2$ in $\mathcal{H}$) are at angle π on the sphere.

i Note

The Bloch sphere contains *all* the information about a qubit and no redundancy, yet it is two-dimensional: a qubit has infinitely many states but a single measurement yields only one bit. Chapter 9 makes this precise (Holevo's bound).

1.6. The Stern–Gerlach experiment

A beam of silver atoms (spin $\frac{1}{2}$) crossing an inhomogeneous magnetic field along z splits into two beams, corresponding to the outcomes $\pm \frac{\hbar}{2}$ of the spin component S_z : a measurement in the basis $\{|\uparrow_z\rangle, |\downarrow_z\rangle\} \equiv \{|0\rangle, |1\rangle\}$.

1. Block the lower beam: the atoms leaving the first magnet are in $|\uparrow_z\rangle$. A second z -magnet sends **all** of them up — repeated measurements agree.
2. Replace the second magnet with one oriented along x , which measures in $\{|\uparrow_x\rangle, |\downarrow_x\rangle\} = \{|+\rangle, |-\rangle\}$: the beam splits 50/50, since $|\langle\pm|0\rangle|^2 = \frac{1}{2}$.
3. Select the $|\uparrow_x\rangle$ atoms and measure z again: the beam splits 50/50. The x -measurement has **erased** the earlier z information — the state collapsed to $|+\rangle$, which has no memory of having been $|0\rangle$.

The experiment shows quantised outcomes, intrinsic randomness, collapse, and the incompatibility of measurements along different axes (Chapter 4).

1.7. Exercises

Exercise 1.1

For each state–basis pair, give the outcome probabilities: (a) $\frac{\sqrt{3}}{2} |0\rangle - \frac{1}{2} |1\rangle$ in $\{|0\rangle, |1\rangle\}$; (b) $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ in $\{|+i\rangle, |-i\rangle\}$; (c) $|+\rangle$ in $\{\frac{\sqrt{3}}{2} |0\rangle - \frac{1}{2} |1\rangle, \frac{1}{2} |0\rangle + \frac{\sqrt{3}}{2} |1\rangle\}$.

Solution

(a) $p_0 = \frac{3}{4}, p_1 = \frac{1}{4}$. (b) With $\langle \pm i | = (\langle 0 | \mp i \langle 1 |) / \sqrt{2}$, $\langle \pm i | \psi \rangle = \frac{1}{2}(1 \pm i)$, so $p_{+i} = p_{-i} = \frac{1}{2}$: $|-\rangle$ lies on the equator at 90° from both $|\pm i\rangle$. (c) $\langle \phi_0 | + \rangle = \frac{1}{\sqrt{2}}(\frac{\sqrt{3}}{2} - \frac{1}{2})$, so $p_0 = \frac{1}{2} \cdot \frac{(\sqrt{3}-1)^2}{4} = \frac{2-\sqrt{3}}{4}$ and $p_1 = \frac{2+\sqrt{3}}{4}$.

Exercise 1.2

Find θ and φ for the states $|-\rangle, |-i\rangle$ and $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4} |1\rangle)$, and locate them on the Bloch sphere.

Solution

All three have $\theta = \pi/2$ (equal weights): they lie on the equator. $|-\rangle$: $\varphi = \pi$ ($-x$ axis). $|-i\rangle$: $\varphi = 3\pi/2$ ($-y$ axis). The third: $\varphi = \pi/4$, halfway between $+x$ and $+y$.

1.8. Sources

Prof. Vallone, *Quantum Information and Computing*, lecture 2 (kets, scalar product, Hilbert space, qubit, Born rule, global phase, collapse, Bloch sphere; Stern–Gerlach homework) and homework 0.1 (exercises 3–4). Nielsen & Chuang, Sections 1.2 and 2.1–2.2. Benenti, Casati & Strini, Vol. I, Chapters 2–3.

2. Operators and observables

Chapter 1 described measurements as choices of an orthonormal basis. Physics usually phrases them differently: one measures a *quantity* — energy, spin along an axis, polarisation — represented by a Hermitian operator whose eigenvalues are the possible outcomes. This chapter develops the operator toolkit (matrix elements, ket-bra notation, projectors, spectral decomposition), proves the expectation-value formula, and studies the Pauli matrices, which generate all qubit observables.

2.1. Linear operators and their matrices

Definition 2.1 (Linear operator). A linear operator $\hat{M}$ on $\mathcal{H}$ maps vectors to vectors, $\hat{M}(a|\psi\rangle + b|\phi\rangle) = a\hat{M}|\psi\rangle + b\hat{M}|\phi\rangle$. In an orthonormal basis $\{|k\rangle\}$ it is fixed by its action on basis vectors,

$$\hat{M}|k\rangle = \sum_{j=0}^{d-1} M_{jk} |j\rangle, \quad M_{jk} = \langle j|\hat{M}|k\rangle.$$

The matrix element follows by taking the scalar product with $\langle j|$: $\langle j|\hat{M}|k\rangle = \sum_{\ell} M_{\ell k} \langle j|\ell\rangle = M_{jk}$. Acting on a generic state, $\hat{M}|\psi\rangle = \sum_{j,k} M_{jk} \psi_k |j\rangle$: the components transform as $\psi'_j = \sum_k M_{jk} \psi_k$, ordinary matrix–vector multiplication.

2.2. Ket-bra notation

The **outer product** $|\phi\rangle\langle\psi|$ is the operator $|\chi\rangle \mapsto |\phi\rangle\langle\psi|\chi\rangle$. Any operator can be written as

$$\hat{M} = \sum_{j,k=0}^{d-1} M_{jk} |j\rangle\langle k|,$$

since applying the right-hand side to $|k\rangle$ returns $\sum_j M_{jk} |j\rangle$. The **adjoint** is defined by $\langle\phi|\hat{M}^\dagger|\psi\rangle = \langle\psi|\hat{M}|\phi\rangle^*$, i.e. $(M^\dagger)_{jk} = M_{kj}^*$, and $(|\phi\rangle\langle\psi|)^\dagger = |\psi\rangle\langle\phi|$.

2.3. Projectors and completeness

Definition 2.2 (Projector). An operator $\hat{P}$ is an (orthogonal) **projector** if $\hat{P}^2 = \hat{P}$ and $\hat{P}^\dagger = \hat{P}$. For a unit vector $|u\rangle$, $\hat{P}_u = |u\rangle\langle u|$ projects onto the line spanned by $|u\rangle$: $\hat{P}_u|\psi\rangle = \langle u|\psi\rangle |u\rangle$.

In the real plane, projecting $\vec{v}$ onto a unit vector $\hat{u}$ gives $(\hat{u} \cdot \vec{v})\hat{u}$ — exactly $|u\rangle\langle u|v\rangle$. The projectors onto the vectors of an orthonormal basis sum to the identity:

$$\sum_{k=0}^{d-1} |k\rangle\langle k| = \hat{\mathbb{1}} \quad (\text{completeness relation}).$$

2. Operators and observables

Inserting this “resolution of the identity” is the single most useful trick in the formalism; for instance it proves the cyclic property of the trace, $\text{Tr}[\hat{A}_1 \cdots \hat{A}_N] = \text{Tr}[\hat{A}_N \hat{A}_1 \cdots \hat{A}_{N-1}]$, where $\text{Tr} \hat{A} = \sum_k \langle k | \hat{A} | k \rangle$.

2.4. Observables and the spectral theorem

Definition 2.3 (Hermitian operator and observable). $\hat{A}$ is **Hermitian** (self-adjoint) if $\hat{A}^\dagger = \hat{A}$. Physical quantities — **observables** — are represented by Hermitian operators.

Theorem 2.1 (Spectral decomposition). *A Hermitian operator has real eigenvalues a and an orthonormal basis of eigenvectors $|a\rangle$, $\hat{A}|a\rangle = a|a\rangle$, so that*

$$\hat{A} = \sum_a a |a\rangle \langle a| = \sum_a a \hat{\Pi}_a,$$

where, for degenerate eigenvalues, $\hat{\Pi}_a$ is the projector onto the whole eigenspace.

Definition 2.4 (Measurement of an observable). Measuring $\hat{A}$ on $|\psi\rangle$ yields one of the eigenvalues a , with probability

$$p_a = \langle \psi | \hat{\Pi}_a | \psi \rangle = |\langle a | \psi \rangle|^2 \text{ (non-degenerate case),}$$

and the state collapses to $\hat{\Pi}_a |\psi\rangle / \sqrt{p_a}$.

Measuring an observable is thus measuring in its eigenbasis and *labelling* each outcome with the eigenvalue. The labels matter for averages.

Proposition 2.2 (Expectation value). *The mean value of the outcomes is*

$$\langle \hat{A} \rangle = \sum_a a p_a = \langle \psi | \hat{A} | \psi \rangle.$$

Proof. $\langle \psi | \hat{A} | \psi \rangle = \sum_a a \langle \psi | a \rangle \langle a | \psi \rangle = \sum_a a |\langle a | \psi \rangle|^2 = \sum_a a p_a$. $\square$

The **variance** is $(\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2$; it vanishes iff $|\psi\rangle$ is an eigenvector of $\hat{A}$.

2.5. Pauli matrices

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

In ket-bra form: $\hat{\sigma}_x = |0\rangle \langle 1| + |1\rangle \langle 0|$ (bit flip), $\hat{\sigma}_y = i|1\rangle \langle 0| - i|0\rangle \langle 1|$, $\hat{\sigma}_z = |0\rangle \langle 0| - |1\rangle \langle 1|$ (phase flip).

Proposition 2.3 (Algebra of Pauli matrices). *1. Each has eigenvalues ± 1 ; the eigenvectors are $|\pm\rangle$ for $\hat{\sigma}_x$, $|\pm i\rangle$ for $\hat{\sigma}_y$, $|0\rangle, |1\rangle$ for $\hat{\sigma}_z$: $\hat{\sigma}_x = |+\rangle \langle +| - |-\rangle \langle -|$, etc.*
2. $\{\hat{\sigma}_j, \hat{\sigma}_k\} = \hat{\sigma}_j \hat{\sigma}_k + \hat{\sigma}_k \hat{\sigma}_j = 2\delta_{jk} \hat{\mathbb{1}}$: they square to the identity and anticommute.
3. $[\hat{\sigma}_j, \hat{\sigma}_k] = 2i\epsilon_{jkl} \hat{\sigma}_l$, so $\hat{\sigma}_j \hat{\sigma}_k = \delta_{jk} \hat{\mathbb{1}} + i\epsilon_{jkl} \hat{\sigma}_l$.
4. $\text{Tr} \hat{\sigma}_j = 0$ and $\text{Tr}[\hat{\sigma}_j \hat{\sigma}_k] = 2\delta_{jk}$.
5. $\{\hat{\mathbb{1}}, \hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$ is a basis of all 2×2 complex matrices; with real coefficients, of all 2×2 Hermitian matrices.

(4) follows from (2) and cyclicity: for $j \neq k$, $\text{Tr}[\hat{\sigma}_j \hat{\sigma}_k] = -\text{Tr}[\hat{\sigma}_k \hat{\sigma}_j] = -\text{Tr}[\hat{\sigma}_j \hat{\sigma}_k]$.

2.6. The generic qubit observable

For a unit vector $\vec{n} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$ define the spin component along $\vec{n}$:

$$\vec{n} \cdot \hat{\sigma} = n_x \hat{\sigma}_x + n_y \hat{\sigma}_y + n_z \hat{\sigma}_z = \begin{pmatrix} \cos \theta & e^{-i\varphi} \sin \theta \\ e^{i\varphi} \sin \theta & -\cos \theta \end{pmatrix}.$$

Proposition 2.4 (Eigen-decomposition of $\vec{n} \cdot \hat{\sigma}$). $(\vec{r} \cdot \hat{\sigma})^2 = |\vec{r}|^2 \mathbb{1}$ for any real $\vec{r}$, so $\vec{n} \cdot \hat{\sigma}$ has eigenvalues ± 1 , with eigenvectors

$$|+\vec{n}\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad |-\vec{n}\rangle = \sin \frac{\theta}{2} |0\rangle - e^{i\varphi} \cos \frac{\theta}{2} |1\rangle.$$

The $+1$ eigenvector of $\vec{n} \cdot \hat{\sigma}$ is precisely the qubit state whose Bloch vector is $\vec{n}$, and the -1 eigenvector the antipodal one. Every qubit observable has the form $a\mathbb{1} + b\vec{n} \cdot \hat{\sigma}$; measuring it is measuring “spin along $\vec{n}$ ”. For a state with Bloch vector $\vec{r}$, $\langle \vec{n} \cdot \hat{\sigma} \rangle = \vec{n} \cdot \vec{r}$; in particular $\langle \hat{\sigma}_x \rangle = r_x$, $\langle \hat{\sigma}_y \rangle = r_y$, $\langle \hat{\sigma}_z \rangle = r_z$ — the Bloch vector *is* the vector of Pauli expectation values.

2.7. Exercises

Exercise 2.1

Show that $\text{Tr}[(\vec{r} \cdot \hat{\sigma})(\vec{s} \cdot \hat{\sigma})] = 2\vec{r} \cdot \vec{s}$ and $(\vec{r} \cdot \hat{\sigma})^2 = |\vec{r}|^2 \mathbb{1}$.

Solution

$(\vec{r} \cdot \hat{\sigma})(\vec{s} \cdot \hat{\sigma}) = \sum_{jk} r_j s_k \hat{\sigma}_j \hat{\sigma}_k = \sum_{jk} r_j s_k (\delta_{jk} \mathbb{1} + i\epsilon_{jkl} \hat{\sigma}_l) = (\vec{r} \cdot \vec{s}) \mathbb{1} + i(\vec{r} \times \vec{s}) \cdot \hat{\sigma}$. Taking the trace kills the Pauli term: $2\vec{r} \cdot \vec{s}$. With $\vec{s} = \vec{r}$ the cross product vanishes: $|\vec{r}|^2 \mathbb{1}$.

Exercise 2.2

Compute $\langle \hat{\sigma}_x \rangle$, $\langle \hat{\sigma}_y \rangle$, $\langle \hat{\sigma}_z \rangle$ for $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$ and check that they are the Cartesian components of the Bloch vector.

Solution

$\langle \hat{\sigma}_z \rangle = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta$. $\langle \hat{\sigma}_x \rangle = 2\text{Re}(\alpha^* \beta) = 2 \cos \frac{\theta}{2} \sin \frac{\theta}{2} \cos \varphi = \sin \theta \cos \varphi$. $\langle \hat{\sigma}_y \rangle = 2\text{Im}(\alpha^* \beta) = \sin \theta \sin \varphi$.

2.8. Sources

Prof. Vallone, *Quantum Information and Computing*, lecture 3 (operators, ket-bra notation, projectors, spectral decomposition, mean values, Pauli matrices, $\vec{n} \cdot \hat{\sigma}$) and homework 0.1 (exercises 1, 2, 5). Nielsen & Chuang, Section 2.1. Benenti, Casati & Strini, Vol. I, Chapter 2.

3. Mixed states and density operators

A ket describes a system prepared in a definite quantum state. Often we know less: a source emits different states at random, or the system is part of a larger entangled whole (Chapter 5). Both situations are described by the **density operator**, which replaces the state vector as the most general description of a quantum state. This chapter defines it, rewrites the Born rule as a trace, distinguishes quantum superpositions from classical mixtures, and extends the Bloch sphere to the Bloch ball.

3.1. Ensembles and the density operator

Suppose a source prepares the state $|\psi_k\rangle$ with probability p_k — an **ensemble** $\{p_k, |\psi_k\rangle\}$. The probability of outcome a in a measurement of $\hat{A}$ is the average of the Born probabilities:

$$p_a = \sum_k p_k \langle \psi_k | \hat{\Pi}_a | \psi_k \rangle = \text{Tr} \left[\hat{\Pi}_a \sum_k p_k |\psi_k\rangle \langle \psi_k| \right],$$

using $\langle \psi | \hat{X} | \psi \rangle = \text{Tr}[\hat{X} |\psi\rangle \langle \psi|]$. All predictions depend on the ensemble only through one operator.

Definition 3.1 (Density operator). The **density operator** of the ensemble $\{p_k, |\psi_k\rangle\}$ is

$$\hat{\rho} = \sum_k p_k |\psi_k\rangle \langle \psi_k|.$$

A state is **pure** if $\hat{\rho} = |\psi\rangle \langle \psi|$ for some $|\psi\rangle$, and **mixed** otherwise.

Theorem 3.1 (Characterisation). An operator $\hat{\rho}$ is a density operator of some ensemble iff it is

1. Hermitian, $\hat{\rho}^\dagger = \hat{\rho}$;
2. positive semidefinite, $\langle \phi | \hat{\rho} | \phi \rangle \geq 0$ for all $|\phi\rangle$;
3. of unit trace, $\text{Tr} \hat{\rho} = 1$.

The trace is basis independent: $\text{Tr} \hat{A} = \sum_k \langle k | \hat{A} | k \rangle$ for any orthonormal basis, and $\text{Tr}(|a\rangle \langle b|) = \langle b | a \rangle$ turns outer products into inner products. Positivity follows from $\langle \phi | \hat{\rho} | \phi \rangle = \sum_k p_k |\langle \phi | \psi_k \rangle|^2$; the trace from normalisation. Conversely, the spectral decomposition $\hat{\rho} = \sum_j \lambda_j |j\rangle \langle j|$ with $\lambda_j \geq 0$, $\sum \lambda_j = 1$ is itself an ensemble. Convex combinations of density operators are density operators: preparing $\hat{\rho}_1$ with probability p_1 and $\hat{\rho}_2$ with p_2 gives $p_1 \hat{\rho}_1 + p_2 \hat{\rho}_2$.

Definition 3.2 (Born rule and expectation values). For a state $\hat{\rho}$:

$$p_a = \text{Tr}[\hat{\Pi}_a \hat{\rho}], \quad \langle \hat{A} \rangle = \text{Tr}[\hat{A} \hat{\rho}], \quad \hat{\rho} \xrightarrow{a} \frac{\hat{\Pi}_a \hat{\rho} \hat{\Pi}_a}{p_a}.$$

3.2. Pure vs. mixed: superposition is not ignorance

Compare $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ with the mixture “ $|0\rangle$ or $|1\rangle$ with probability $\frac{1}{2}$ ”:

$$\hat{\rho}_{\text{pure}} = |+\rangle\langle+| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \hat{\rho}_{\text{mix}} = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

In the computational basis both give $p_0 = p_1 = \frac{1}{2}$. In the $\{| \pm \rangle\}$ basis, the pure state gives $p_+ = 1$, the mixture still $\frac{1}{2}$. The **off-diagonal elements** (coherences) carry the phase information that distinguishes a superposition from a classical coin flip.

Definition 3.3 (Purity). $\text{Tr} \hat{\rho}^2 = \sum_j \lambda_j^2 \leq 1$, with equality iff $\hat{\rho}$ is pure. The minimum $1/d$ is reached by the **maximally mixed state** $\hat{\mathbb{1}}/d$.

3.3. The Bloch ball

Since $\{\hat{\mathbb{1}}, \hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$ is a basis of Hermitian 2×2 matrices and $\text{Tr} \hat{\rho} = 1$,

$$\hat{\rho} = \frac{1}{2}(\hat{\mathbb{1}} + \vec{r} \cdot \hat{\vec{\sigma}}) = \frac{1}{2} \begin{pmatrix} 1 + r_z & r_x - ir_y \\ r_x + ir_y & 1 - r_z \end{pmatrix}, \quad r_j = \text{Tr}[\hat{\sigma}_j \hat{\rho}] = \langle \hat{\sigma}_j \rangle.$$

Proposition 3.2 (Bloch ball). *The eigenvalues of $\hat{\rho}$ are $\lambda_{\pm} = \frac{1}{2}(1 \pm |\vec{r}|)$, with eigenvectors $|\pm_{\vec{r}}\rangle$ (the pure states along $\pm \vec{r}$). Hence $\hat{\rho} \geq 0$ iff $|\vec{r}| \leq 1$: qubit states fill the unit **ball**. Pure states are on the surface ($|\vec{r}| = 1$, $\text{Tr} \hat{\rho}^2 = \frac{1}{2}(1 + |\vec{r}|^2) = 1$); the maximally mixed state is the centre.*

Every mixed state is a convex combination of the two pure states along its axis: $\hat{\rho} = \lambda_+ |+_{\vec{r}}\rangle\langle+_{\vec{r}}| + \lambda_- |-_{\vec{r}}\rangle\langle-_{\vec{r}}|$. Geometrically, mixing pulls the Bloch vector inside the sphere.

Proposition 3.3 (Overlap of pure states). *For pure states $\hat{\rho}_1 = \frac{1}{2}(\hat{\mathbb{1}} + \vec{r} \cdot \hat{\vec{\sigma}})$, $\hat{\rho}_2 = \frac{1}{2}(\hat{\mathbb{1}} + \vec{s} \cdot \hat{\vec{\sigma}})$,*

$$|\langle \psi | \phi \rangle|^2 = \text{Tr}[\hat{\rho}_1 \hat{\rho}_2] = \frac{1 + \vec{r} \cdot \vec{s}}{2}.$$

Two pure states are orthogonal iff $\vec{r} \cdot \vec{s} = -1$, i.e. iff they are antipodal.

Proof. Expand the product and use $\text{Tr} \hat{\sigma}_j = 0$, $\text{Tr}[\hat{\sigma}_j \hat{\sigma}_k] = 2\delta_{jk}$: $\frac{1}{4}(2 + 2\vec{r} \cdot \vec{s})$. $\square$

3.4. Many ensembles, one density matrix

The map from ensembles to density operators is many-to-one, and **no measurement can tell the ensembles apart**.

Example 3.1 (*Two indistinguishable sources*)

Source A emits $|0\rangle$ with probability $\frac{3}{4}$ and $|1\rangle$ with probability $\frac{1}{4}$. Source B emits $|\phi_{\pm}\rangle = \frac{\sqrt{3}}{2}|0\rangle \pm \frac{1}{2}|1\rangle$ with probability $\frac{1}{2}$ each. Then

$$\hat{\rho}_B = \frac{1}{2} \sum_{\pm} |\phi_{\pm}\rangle \langle \phi_{\pm}| = \frac{3}{4} |0\rangle \langle 0| + \frac{1}{4} |1\rangle \langle 1| = \hat{\rho}_A,$$

because the cross terms $\pm \frac{\sqrt{3}}{4}(|0\rangle \langle 1| + |1\rangle \langle 0|)$ cancel. On the Bloch ball, $\vec{r} = (0, 0, \frac{1}{2})$ is the midpoint of the segment joining the poles with weights $\frac{3}{4}, \frac{1}{4}$, and also of the segment joining the Bloch vectors of $|\phi_{\pm}\rangle$.

In general, $\{p_k, |\psi_k\rangle\}$ and $\{q_j, |\phi_j\rangle\}$ give the same $\hat{\rho}$ iff $\sqrt{p_k} |\psi_k\rangle = \sum_j u_{kj} \sqrt{q_j} |\phi_j\rangle$ for some unitary matrix u (Hughston–Jozsa–Wootters). The density operator is everything that is physically meaningful about a preparation.

! Important

Do not read $\hat{\rho}$ as “the system is really in one of the $|\psi_k\rangle$ and we don’t know which”. The same $\hat{\rho}$ arises from incompatible stories. For a subsystem of an entangled state (Chapter 5) there is no such story at all.

3.5. Exercises**Exercise 3.1**

A qubit has Bloch vector $\vec{r} = r(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, $r < 1$. Compute the outcome probabilities when measuring $\hat{\sigma}_z$ and $\hat{\sigma}_x$.

Solution

$p_{0,1} = \frac{1}{2}(1 \pm r_z) = \frac{1}{2}(1 \pm r/\sqrt{2})$; $p_{\pm} = \frac{1}{2}(1 \pm r_x) = \frac{1}{2}$. (Measuring $\hat{\sigma}_y$ would give $\frac{1}{2}(1 \pm r/\sqrt{2})$.)

Exercise 3.2

Give an example of a Hermitian operator that is not positive, and show that a convex combination of density operators is a density operator.

Solution

Any Pauli matrix, e.g. $\hat{\sigma}_z$ with eigenvalue -1 . For $\hat{\rho} = p_1 \hat{\rho}_1 + p_2 \hat{\rho}_2$: $\text{Tr } \hat{\rho} = p_1 + p_2 = 1$; $\hat{\rho}^\dagger = \hat{\rho}$ since p_i are real; $\langle \phi | \hat{\rho} | \phi \rangle = p_1 \langle \phi | \hat{\rho}_1 | \phi \rangle + p_2 \langle \phi | \hat{\rho}_2 | \phi \rangle \geq 0$. It is prepared by tossing a biased coin and preparing $\hat{\rho}_1$ or $\hat{\rho}_2$.

3.6. Sources

Prof. Vallone, *Quantum Information and Computing*, lecture 4 (ensembles, density operator, trace, Born rule as trace, pure vs. mixed, Bloch ball, equivalent ensembles) and homework 0.2 (exercises 1 and 3). Nielsen & Chuang, Section 2.4. Benenti, Casati & Strini, Vol. I, Chapter 5, and Vol. II, Chapter 5 (suggested in the lecture).

4. Unitary evolution and the postulates

States change in two ways: by measurement, which is random and irreversible, and by the deterministic, reversible evolution of an isolated system, described by **unitary operators**. This chapter studies unitaries — in particular single-qubit rotations — derives them from the Schrödinger equation, discusses the incompatibility of measurements and the uncertainty relation, and collects the postulates of quantum mechanics for isolated systems.

4.1. Unitary operators

Definition 4.1 (Unitary operator). $\hat{U}$ is **unitary** if $\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{1}$. It preserves scalar products, $\langle \hat{U}\phi | \hat{U}\psi \rangle = \langle \phi | \psi \rangle$, hence norms and orthogonality; it maps orthonormal bases to orthonormal bases; its eigenvalues have unit modulus, $e^{i\lambda}$.

A unitary transforms states as

$$|\psi\rangle \mapsto \hat{U} |\psi\rangle, \quad \hat{\rho} \mapsto \hat{U} \hat{\rho} \hat{U}^\dagger.$$

Because $\hat{U}$ is invertible, the transformation is reversible: no information is lost.

Warning

Three kinds of operators play very different roles and should not be confused:

- **observables** $\hat{A} = \hat{A}^\dagger$ label measurement outcomes; *applying* $\hat{A}$ to a state, $\hat{A} |\psi\rangle$, has no physical meaning (it is not even normalised);
- **density operators** $\hat{\rho} \geq 0$, $\text{Tr} \hat{\rho} = 1$ describe states;
- **unitaries** $\hat{U}^\dagger \hat{U} = \hat{1}$ describe transformations.

The Pauli matrices are special: they are both Hermitian and unitary, so $\hat{\sigma}_x$ is at the same time “spin along x ” (an observable) and “bit flip” (a transformation). Projectors $\hat{\Pi}_a$ appear inside measurements, but “ $\hat{\Pi}_a |\psi\rangle$ ” is only the unnormalised post-measurement state.

4.2. Single-qubit unitaries

Every 2×2 unitary can be written as

$$\hat{U} = e^{i\alpha} \hat{R}_{\vec{n}}(\theta), \quad \hat{R}_{\vec{n}}(\theta) = e^{-i\frac{\theta}{2} \vec{n} \cdot \hat{\vec{\sigma}}} = \cos \frac{\theta}{2} \hat{1} - i \sin \frac{\theta}{2} \vec{n} \cdot \hat{\vec{\sigma}},$$

where the closed form follows from $(\vec{n} \cdot \hat{\vec{\sigma}})^2 = \hat{1}$. The global phase $e^{i\alpha}$ is unobservable, and $\hat{R}_{\vec{n}}(\theta)$ acts on the Bloch sphere as a **rotation by angle θ about the axis $\vec{n}$** . The “half angle” $\theta/2$ is the same doubling met in Chapter 1: $\hat{R}_{\vec{n}}(2\pi) = -\hat{1}$, a global phase.

Equivalently, any single-qubit unitary decomposes (Euler) as $\hat{U} = e^{i\alpha} \hat{R}_z(\beta) \hat{R}_y(\gamma) \hat{R}_z(\delta)$. Familiar examples:

4. Unitary evolution and the postulates

$\hat{\sigma}_x = i\hat{R}_x(\pi)$ (bit flip, rotation by π about x), $\hat{\sigma}_z = i\hat{R}_z(\pi)$ (phase flip), and the **Hadamard** gate

$$\hat{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{\hat{\sigma}_x + \hat{\sigma}_z}{\sqrt{2}},$$

a rotation by π about the axis $(\hat{x} + \hat{z})/\sqrt{2}$, which swaps $|0\rangle \leftrightarrow |+\rangle$ and $|1\rangle \leftrightarrow |-\rangle$, i.e. exchanges the z and x bases.

4.3. Time evolution

Definition 4.2 (Schrödinger equation). The state of an isolated system evolves according to

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

where the Hamiltonian $\hat{H}$ is the observable associated with energy.

For a time-independent Hamiltonian the solution is

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle, \quad \hat{U}(t) = e^{-i\hat{H}t/\hbar} = \sum_n e^{-iE_n t/\hbar} |E_n\rangle \langle E_n|,$$

with $\hat{H} |E_n\rangle = E_n |E_n\rangle$. Each energy eigenstate only acquires a phase (it is *stationary*); superpositions evolve because the relative phases change. Since $\hat{H}$ is Hermitian, $\hat{U}(t)$ is unitary. In density-operator language, $i\hbar \dot{\hat{\rho}} = [\hat{H}, \hat{\rho}]$ (von Neumann equation).

Example 4.1 (Larmor precession)

Let $\hat{H} = B_0 \hat{\sigma}_z$ (a spin in a magnetic field along z , with constants absorbed in B_0) and $|\psi(0)\rangle = |+\rangle$. Then

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} (e^{-iB_0 t/\hbar} |0\rangle + e^{iB_0 t/\hbar} |1\rangle) \simeq \frac{1}{\sqrt{2}} (|0\rangle + e^{i\omega t} |1\rangle), \quad \omega = \frac{2B_0}{\hbar}.$$

The Bloch vector rotates around z at angular frequency ω : $\langle \hat{\sigma}_x \rangle = \cos \omega t$, $\langle \hat{\sigma}_y \rangle = \sin \omega t$, $\langle \hat{\sigma}_z \rangle = 0$. Indeed $\hat{U}(t) = \hat{R}_z(\omega t)$. Every single-qubit gate is realised physically by switching on a suitable Hamiltonian for a suitable time.

4.4. Incompatible measurements

Two observables $\hat{A}, \hat{B}$ can be measured “simultaneously” — have a common eigenbasis, so that measuring one does not disturb the statistics of the other — iff they **commute**, $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = 0$. Otherwise they are **incompatible**: a state cannot have sharp values of both.

Theorem 4.1 (Robertson uncertainty relation). For any state and any observables $\hat{A}, \hat{B}$,

$$\Delta A \Delta B \geq \frac{1}{2} |[\hat{A}, \hat{B}]|.$$

Proof. Let $\hat{a} = \hat{A} - \langle \hat{A} \rangle$, $\hat{b} = \hat{B} - \langle \hat{B} \rangle$. By Cauchy-Schwarz, $(\Delta A)^2 (\Delta B)^2 = \langle \hat{a}^2 \rangle \langle \hat{b}^2 \rangle \geq |\langle \hat{a}\hat{b} \rangle|^2 \geq |\text{Im} \langle \hat{a}\hat{b} \rangle|^2 = \frac{1}{4} |[\hat{A}, \hat{B}]|^2$, since $\hat{a}\hat{b} - \hat{b}\hat{a} = [\hat{A}, \hat{B}]$ and $\langle \hat{a}\hat{b} \rangle^* = \langle \hat{b}\hat{a} \rangle$. $\square$

Example 4.2 ($\hat{\sigma}_x$ and $\hat{\sigma}_z$)

$[\hat{\sigma}_z, \hat{\sigma}_x] = 2i\hat{\sigma}_y$, so $\Delta\sigma_z \Delta\sigma_x \geq |\langle \hat{\sigma}_y \rangle|$. On $|0\rangle$, $\Delta\sigma_z = 0$ and $\Delta\sigma_x = 1$: definite in z , maximally uncertain in x — the Stern–Gerlach result of Chapter 1. For pure qubit states $(\Delta\sigma_j)^2 = 1 - r_j^2$, and the relation reads $(1 - r_z^2)(1 - r_x^2) \geq r_y^2$, true on the sphere.

Incompatibility is a quantum resource: it is what makes eavesdropping detectable in quantum cryptography (Chapter 10) and what Bell’s theorem turns into a proof of nonlocality (Chapter 6). The “quantum cakes” of Chapter 6 are a story about two incompatible properties.

4.5. The postulates for isolated systems

1. **States.** The state of an isolated system is a unit vector $|\psi\rangle$ in a Hilbert space $\mathcal{H}$, defined up to a global phase (more generally a density operator $\hat{\rho}$). Any normalised superposition of states is a state.
2. **Evolution.** States evolve according to the Schrödinger equation, $i\hbar \partial_t |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$; for time-independent $\hat{H}$, $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle$. Any discrete transformation is a unitary.
3. **Measurement.** Observables are Hermitian operators $\hat{A} = \sum_a a \hat{\Pi}_a$. The possible outcomes are the eigenvalues a , obtained with probability $p_a = |\langle a|\psi\rangle|^2 = \text{Tr}[\hat{\Pi}_a \hat{\rho}]$.
4. **Collapse.** After obtaining a , the state becomes $\hat{\Pi}_a |\psi\rangle / \sqrt{p_a}$.
5. **Composite systems.** The state space of a composite system is the tensor product of the components’ spaces (Chapter 5).

Chapter 8 generalises postulates 2–4 to systems that interact with an environment.

4.6. Exercises

Exercise 4.1

Show that $\hat{H}\hat{\sigma}_x\hat{H} = \hat{\sigma}_z$, $\hat{H}\hat{\sigma}_z\hat{H} = \hat{\sigma}_x$ and $\hat{H}\hat{\sigma}_y\hat{H} = -\hat{\sigma}_y$. Interpret geometrically.

Solution

Direct multiplication, or use $\hat{H} = (\hat{\sigma}_x + \hat{\sigma}_z)/\sqrt{2}$ and the anticommutation rules: $\hat{H}\hat{\sigma}_x\hat{H} = \frac{1}{2}(\hat{\sigma}_x + \hat{\sigma}_z)\hat{\sigma}_x(\hat{\sigma}_x + \hat{\sigma}_z) = \frac{1}{2}(\hat{\sigma}_x + \hat{\sigma}_z + \hat{\sigma}_z\hat{\sigma}_x\hat{\sigma}_z + \hat{\sigma}_x)$, and $\hat{\sigma}_z\hat{\sigma}_x\hat{\sigma}_z = -\hat{\sigma}_x$, giving $\hat{\sigma}_z$. Geometrically $\hat{H}$ is a π rotation about $(\hat{x} + \hat{z})/\sqrt{2}$, which swaps the x and z axes and reverses y .

Exercise 4.2

A qubit starts in $|0\rangle$ and evolves under $\hat{H} = \frac{\hbar\Omega}{2}\hat{\sigma}_x$. Find the probability of finding it in $|1\rangle$ at time t .

Solution

$\hat{U}(t) = e^{-i\Omega t \hat{\sigma}_x/2} = \hat{R}_x(\Omega t)$, so $|\psi(t)\rangle = \cos \frac{\Omega t}{2} |0\rangle - i \sin \frac{\Omega t}{2} |1\rangle$ and $p_1(t) = \sin^2 \frac{\Omega t}{2}$: Rabi oscillations. A pulse with $\Omega t = \pi$ is a bit flip (up to a phase).

4.7. Sources

Prof. Vallone, *Quantum Information and Computing*, lectures 5 (unitaries, single-qubit unitaries, observables vs. density matrices vs. unitaries, incompatible measurements, uncertainty principle) and 9 (Schrödinger

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evolution, single-qubit rotations, the $\hat{H} = B_0 \hat{\sigma}_z$ exercise, postulates). Nielsen & Chuang, Sections 2.1.7–2.2 and 4.2. Benenti, Casati & Strini, Vol. I, Chapters 2–3.

5. Composite systems and entanglement

Quantum information becomes genuinely new when several systems are considered together. The state space of a composite system is the **tensor product** of its parts, which contains states that cannot be described part by part: **entangled** states. This chapter builds the tensor product and the partial trace, defines entanglement and the Bell states, studies the perfect correlations of the singlet, and introduces the Schmidt decomposition, which settles whether a pure bipartite state is entangled.

5.1. Tensor products

Think of two “quantum balls”, each of which can be red or blue. Classically the pair is in one of the four states rr, rb, br, bb ; quantum mechanically the pair can be in any superposition of them.

Definition 5.1 (Tensor product of Hilbert spaces). If $\mathcal{H}_A$ has orthonormal basis $\{|a_j\rangle\}_{j=1}^m$ and $\mathcal{H}_B$ has $\{|b_k\rangle\}_{k=1}^n$, the tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$ is the mn -dimensional space spanned by the vectors $|a_j\rangle \otimes |b_k\rangle \equiv |a_j, b_k\rangle \equiv |a_j\rangle |b_k\rangle$. The tensor product is bilinear,

$$(|a_1\rangle + |a_2\rangle) \otimes (|b_1\rangle + |b_2\rangle) = |a_1 b_1\rangle + |a_1 b_2\rangle + |a_2 b_1\rangle + |a_2 b_2\rangle,$$

and the scalar product factorises on product vectors: $\langle a_1 b_1 | a_2 b_2 \rangle = \langle a_1 | a_2 \rangle \langle b_1 | b_2 \rangle$.

A generic state of two qubits is $|\Psi\rangle_{AB} = \sum_{a,b \in \{0,1\}} \psi_{ab} |ab\rangle$ — four complex amplitudes, not the two pairs of amplitudes a description “qubit by qubit” would give. For N qubits the dimension is 2^N : the exponential growth that makes quantum systems hard to simulate classically and quantum computers potentially powerful.

Definition 5.2 (Tensor product of operators). $(\hat{A} \otimes \hat{B}) |a_j, b_k\rangle = (\hat{A} |a_j\rangle) \otimes (\hat{B} |b_k\rangle)$, extended by linearity. In matrix form this is the **Kronecker product**: the block matrix $[A_{jj'} B]$.

Example 5.1 (Kronecker product)

$$\hat{\sigma}_x \otimes \hat{\sigma}_z = \begin{pmatrix} 0 \cdot \hat{\sigma}_z & 1 \cdot \hat{\sigma}_z \\ 1 \cdot \hat{\sigma}_z & 0 \cdot \hat{\sigma}_z \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

in the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Useful rules: $(\hat{A} \otimes \hat{B})(\hat{C} \otimes \hat{D}) = \hat{A}\hat{C} \otimes \hat{B}\hat{D}$, $(\hat{A} \otimes \hat{B})^\dagger = \hat{A}^\dagger \otimes \hat{B}^\dagger$, $\text{Tr}[\hat{A} \otimes \hat{B}] = \text{Tr}\hat{A} \text{Tr}\hat{B}$. A local operation on A alone is $\hat{A} \otimes \hat{1}_B$.

5.2. Partial trace and reduced states

Alice holds A and Bob holds B of a joint state $|\psi\rangle_{AB}$. What does Alice see if she measures in a basis $\{|a\rangle\}$ and ignores Bob? Summing the joint probabilities over Bob's outcomes in any basis $\{|b\rangle\}$,

$$P(a) = \sum_b P(a, b) = \sum_b |\langle a| \otimes \langle b| |\psi\rangle|^2 = \sum_b (\langle a| \otimes \langle b| |\psi\rangle \langle \psi| (|a\rangle \otimes |b\rangle)) = \langle a| \text{Tr}_B[|\psi\rangle \langle \psi|] |a\rangle.$$

Definition 5.3 (Partial trace and reduced density operator). For an operator $\hat{X}$ on $\mathcal{H}_A \otimes \mathcal{H}_B$,

$$\text{Tr}_B \hat{X} = \sum_{b=0}^{d_B-1} (\hat{1}_A \otimes \langle b|) \hat{X} (\hat{1}_A \otimes |b\rangle),$$

equivalently $\text{Tr}_B[|a\rangle \langle a'| \otimes |b\rangle \langle b'|] = |a\rangle \langle a'| \langle b'|b\rangle$. The **reduced density operator** of A is $\hat{\rho}_A = \text{Tr}_B \hat{\rho}_{AB}$; its matrix elements are $\langle a|\hat{\rho}_A|a'\rangle = \sum_b \langle a, b|\hat{\rho}_{AB}|a', b\rangle$.

$\hat{\rho}_A$ is a genuine density operator (Hermitian, positive, unit trace — positivity because $\langle \phi|\hat{\rho}_A|\phi\rangle = \sum_b \langle \phi, b|\hat{\rho}_{AB}|\phi, b\rangle \geq 0$), and it predicts *every* local measurement on A : $\text{Tr}[(\hat{O}_A \otimes \hat{1})\hat{\rho}_{AB}] = \text{Tr}[\hat{O}_A \hat{\rho}_A]$. It is the unique such operator. The construction generalises to N subsystems by tracing out all but the ones of interest.

5.3. Entanglement

Definition 5.4 (Product and entangled pure states). A pure state $|\psi\rangle_{AB}$ is a **product state** if $|\psi\rangle_{AB} = |\alpha\rangle_A \otimes |\beta\rangle_B$; otherwise it is **entangled**.

Definition 5.5 (Bell states). The four maximally entangled two-qubit states

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}$$

form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$, the **Bell basis**. $|\Psi^-\rangle$ is the **singlet**.

For every Bell state $\hat{\rho}_A = \hat{\rho}_B = \hat{1}/2$: each qubit alone is maximally mixed, although the pair is in a pure state. **The whole is known perfectly; the parts are completely random.** This cannot happen classically, where knowing a joint distribution with certainty means knowing each marginal with certainty.

Correlations in the singlet. If Alice and Bob both measure $\hat{\sigma}_z$ on $|\Psi^-\rangle$, each gets a random result, but the results are always opposite. Rewriting the state in the x basis,

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} = \frac{|-+\rangle - |+-\rangle}{\sqrt{2}},$$

so $\hat{\sigma}_x$ measurements are also perfectly anticorrelated — and in fact along any common axis $\vec{n}$, because $(\hat{U} \otimes \hat{U}) |\Psi^-\rangle = \det(\hat{U}) |\Psi^-\rangle$ for every unitary $\hat{U}$: the singlet is rotationally invariant. In general $\langle (\vec{a} \cdot \hat{\sigma}) \otimes (\vec{b} \cdot \hat{\sigma}) \rangle_{\Psi^-} = -\vec{a} \cdot \vec{b}$.

A classical source could produce anticorrelated pairs in one fixed basis (red/blue socks), but not simultaneously in incompatible bases. Chapter 6 shows that no local “instruction set” can reproduce all the singlet correlations.

Remark (No signalling). When Alice measures, Bob's conditional state collapses instantaneously. Yet Bob's *reduced* state $\hat{\rho}_B = \text{Tr}_A \hat{\rho}_{AB}$ is unaffected by anything Alice does locally (unitaries or measurements whose outcomes Bob does not know): $\text{Tr}_A[(\hat{U}_A \otimes \hat{1})\hat{\rho}_{AB}(\hat{U}_A^\dagger \otimes \hat{1})] = \hat{\rho}_B$ by cyclicity of the partial trace over A . Entanglement produces correlations but cannot transmit information faster than light.

5.4. Schmidt decomposition

Theorem 5.1 (Schmidt decomposition). *Every pure state of $\mathcal{H}_A \otimes \mathcal{H}_B$ ($d_A \leq d_B$) can be written as*

$$|\psi\rangle_{AB} = \sum_{k=0}^{d_A-1} \sqrt{p_k} |v_k\rangle_A |w_k\rangle_B,$$

with $\{|v_k\rangle\}$ and $\{|w_k\rangle\}$ orthonormal sets and $p_k \geq 0$, $\sum_k p_k = 1$. The p_k are the eigenvalues of both $\hat{\rho}_A$ and $\hat{\rho}_B$, and $|v_k\rangle$ are the eigenvectors of $\hat{\rho}_A$.

Proof. Expand $|\psi\rangle = \sum_a |v_a\rangle \otimes |\tilde{w}_a\rangle$ in the eigenbasis $\{|v_a\rangle\}$ of $\hat{\rho}_A$, where $|\tilde{w}_a\rangle = (\langle v_a| \otimes \hat{1})|\psi\rangle$ are unnormalised. Then $\hat{\rho}_A = \sum_{a,a'} \langle \tilde{w}_{a'} | \tilde{w}_a \rangle |v_a\rangle \langle v_{a'}|$; since $\hat{\rho}_A$ is diagonal in this basis with eigenvalues p_a , $\langle \tilde{w}_{a'} | \tilde{w}_a \rangle = p_a \delta_{aa'}$. Setting $|w_a\rangle = |\tilde{w}_a\rangle / \sqrt{p_a}$ (for $p_a > 0$) gives orthonormal vectors. $\square$

The number of non-zero p_k is the **Schmidt rank**.

Corollary 5.2 (Entanglement criterion for pure states). *A pure state $|\psi\rangle_{AB}$ is entangled iff its Schmidt rank is larger than 1, iff its reduced state $\hat{\rho}_A$ is mixed ($\text{Tr} \hat{\rho}_A^2 < 1$).*

Local unitaries change the Schmidt bases but not the coefficients: the p_k characterise the entanglement of a pure state completely. Chapter 9 quantifies it with the von Neumann entropy of $\hat{\rho}_A$.

5.5. Exercises

Exercise 5.1

Two sources emit pairs of qubits in $|\psi_1\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$ and $|\psi_2\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$. (a) Compute the reduced states; which pair is entangled? (b) Given only the first qubit of each pair, how can you tell the sources apart? (c) What if, for each pair, you receive the first or the second qubit at random?

Solution

(a) $|\psi_1\rangle = |+\rangle|-\rangle$ is a product: $\hat{\rho}_{1,A} = |+\rangle\langle+|$, $\hat{\rho}_{1,B} = |-\rangle\langle-|$. $|\psi_2\rangle = (|0\rangle|+\rangle + |1\rangle|-\rangle)/\sqrt{2}$ is entangled (Schmidt rank 2): $\hat{\rho}_{2,A} = \hat{\rho}_{2,B} = \hat{1}/2$. (b) Measure in $\{|\pm\rangle\}$: source 1 always gives $+$, source 2 gives $\pm$ at random. (c) Source 1 now delivers $\frac{1}{2}|+\rangle\langle+| + \frac{1}{2}|-\rangle\langle-| = \hat{1}/2$, the same as source 2: the sources are indistinguishable.

Exercise 5.2

For $|\phi\rangle_{AB} = \frac{1}{\sqrt{2}}[|00\rangle + \cos 2\theta |10\rangle + \sin 2\theta |11\rangle]$, $0 \leq \theta \leq \pi/2$: compute $\hat{\rho}_A$, its spectrum, decide whether the state is entangled, and find its Schmidt decomposition.

Solution

With ψ_{ab} arranged as a matrix (rows a , columns b), $\hat{\rho}_A = \psi\psi^\dagger = \frac{1}{2} \begin{pmatrix} 1 & \cos 2\theta \\ \cos 2\theta & 1 \end{pmatrix}$: eigenvalues $\frac{1}{2}(1 \pm \cos 2\theta) = \cos^2 \theta, \sin^2 \theta$ with eigenvectors $|\pm\rangle$. Entangled unless $\theta = 0$ or $\pi/2$. Projecting, $(\langle + | \otimes \hat{1}) |\phi\rangle = \cos \theta (\cos \theta |0\rangle + \sin \theta |1\rangle)$ and $(\langle - | \otimes \hat{1}) |\phi\rangle = \sin \theta (\sin \theta |0\rangle - \cos \theta |1\rangle)$, so

$$|\phi\rangle = \cos \theta |+\rangle (\cos \theta |0\rangle + \sin \theta |1\rangle) + \sin \theta |-\rangle (\sin \theta |0\rangle - \cos \theta |1\rangle).$$

Maximal entanglement occurs at $\theta = \pi/4$, where $|\phi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.

5.6. Sources

Prof. Vallone, *Quantum Information and Computing*, lectures 5–6 (tensor product, partial trace, reduced density matrix, entangled and Bell states, singlet correlations), homework 0.2 (exercises 2–3) and the evaluated homework (exercise 1, Schmidt decomposition). Nielsen & Chuang, Sections 2.1.7, 2.4.3 and 2.5. Benenti, Casati & Strini, Vol. I, Chapters 2 and 5.

6. Nonlocality and Bell inequalities

Entangled states produce correlations between distant measurements. Are they merely the correlations of a pair of socks — determined at the source and revealed later — or something no classical mechanism can mimic? Einstein, Podolsky and Rosen argued for the first view; Bell showed that it makes testable predictions that quantum mechanics violates. This chapter starts from a vivid paradox (the “quantum cakes”), then states the EPR argument, local hidden variables, and the CHSH inequality with its quantum violation.

6.1. The quantum cakes

Two ovens on a conveyor belt bake one cake each, for Alice and Bob. On each cake one can test two **incompatible** properties: open the oven midway and see whether the cake has **R**isen early or **N**ot (which ruins the cooking), or wait until the end and taste whether it is **G**ood or **B**ad. Each tester chooses one of the two tests, at random. After many repetitions:

1. when both check midway, both cakes have risen early 9% of the time;
2. whenever Alice’s cake rose early, Bob’s tasted good;
3. whenever Bob’s cake rose early, Alice’s tasted good.

How often should both cakes taste good when both wait? Classical reasoning: in the 9% of runs where both cakes rise, the properties “would” also make both good — by (2) and (3) — so $P(G_A, G_B) \geq 9\%$. Quantum mechanically one can have $P(G_A, G_B) = 0$.

Example 6.1 (*The cake state*)

Encode $|G\rangle = |0\rangle$, $|B\rangle = |1\rangle$ and the midway test in the rotated basis

$$|R\rangle = \sqrt{\frac{2}{5}}|0\rangle + \sqrt{\frac{3}{5}}|1\rangle, \quad |N\rangle = \sqrt{\frac{3}{5}}|0\rangle - \sqrt{\frac{2}{5}}|1\rangle,$$

and prepare

$$|\psi\rangle = \frac{1}{2}|11\rangle - \sqrt{\frac{3}{8}}(|01\rangle + |10\rangle).$$

Then $\langle GG|\psi\rangle = 0$, so $P(G_A, G_B) = 0$; $\langle RB|\psi\rangle = \frac{1}{2}\sqrt{\frac{3}{5}} - \sqrt{\frac{3}{8}}\sqrt{\frac{2}{5}} = 0$, so a risen cake on one side implies a good one on the other; and $\langle RR|\psi\rangle = \frac{1}{2} \cdot \frac{3}{5} - 2\sqrt{\frac{3}{8}} \cdot \frac{\sqrt{6}}{5} = -0.3$, so $P(R_A, R_B) = 0.09$.

The paradox (Hardy, 1993; popularised by Kwiat and Hardy, 2000) disappears only if we give up assigning definite values to properties that were not measured. “Taste” and “rising” are incompatible, and there is no fact of the matter about the taste of a cake that was opened midway.

6.2. EPR and local hidden variables

Einstein, Podolsky and Rosen (1935) considered a maximally entangled pair. If Alice measures $\hat{\sigma}_z$ she can predict with certainty Bob’s $\hat{\sigma}_z$ result; if she measures $\hat{\sigma}_x$, Bob’s $\hat{\sigma}_x$ result. Assuming

6. Nonlocality and Bell inequalities

- **realism:** if the value of a quantity can be predicted with certainty without disturbing the system, it corresponds to an *element of reality*;
- **locality:** Alice's choice cannot disturb Bob's distant particle,

Bob's particle must carry definite values of both σ_z and σ_x — which quantum mechanics, forbidding simultaneous eigenstates, does not describe. EPR concluded that quantum mechanics is **incomplete**.

The natural completion is a **local hidden-variable (LHV)** model: the source emits a hidden variable λ with distribution $\rho(\lambda)$; Alice's outcome $A(a, \lambda) = \pm 1$ depends only on her setting a and λ , Bob's $B(b, \lambda) = \pm 1$ only on b and λ . Correlations are

$$E(a, b) = \int d\lambda \rho(\lambda) A(a, \lambda) B(b, \lambda).$$

Such models reproduce perfect anticorrelation in any *single* basis. Bell (1964) showed they cannot reproduce all quantum correlations.

6.3. The CHSH inequality

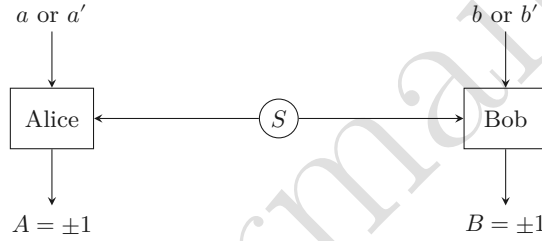


Figure: a Bell test. A source distributes pairs; each party chooses one of two settings at random and records a ± 1 outcome.

Theorem 6.1 (CHSH inequality (Clauser, Horne, Shimony, Holt, 1969)). *For every local hidden-variable model,*

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b') \quad \text{satisfies} \quad |S| \leq 2.$$

Proof. For fixed λ , write $A = A(a, \lambda)$, $A' = A(a', \lambda)$, $B = B(b, \lambda)$, $B' = B(b', \lambda)$, all ± 1 . Then

$$AB + AB' + A'B - A'B' = A(B + B') + A'(B - B').$$

Since $B, B' = \pm 1$, one of $B + B'$, $B - B'$ is 0 and the other ± 2 ; so the expression is ± 2 . Averaging over $\rho(\lambda)$ gives $|S| \leq 2$. $\square$

The proof uses only that all four outcomes A, A', B, B' *exist* for each run (realism) and that A does not depend on b (locality).

Proposition 6.2 (Quantum violation). *For the singlet, $E(\vec{a}, \vec{b}) = \langle (\vec{a} \cdot \hat{\sigma}) \otimes (\vec{b} \cdot \hat{\sigma}) \rangle = -\vec{a} \cdot \vec{b}$. With coplanar settings $\vec{a} = \hat{z}$, $\vec{a}' = \hat{x}$, $\vec{b} = (\hat{z} + \hat{x})/\sqrt{2}$, $\vec{b}' = (\hat{z} - \hat{x})/\sqrt{2}$,*

$$S = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -2\sqrt{2}, \quad |S| = 2\sqrt{2} > 2.$$

Theorem 6.3 (Tsirelson bound). *For any quantum state and any ± 1 -valued observables, $|S| \leq 2\sqrt{2}$.*

Proof sketch. With $\hat{C} = \hat{A} \otimes (\hat{B} + \hat{B}') + \hat{A}' \otimes (\hat{B} - \hat{B}')$ and $\hat{A}^2 = \hat{B}^2 = \hat{1}$, one computes $\hat{C}^2 = 4\hat{1} - [\hat{A}, \hat{A}'] \otimes [\hat{B}, \hat{B}']$, whose norm is at most 8; hence $|\langle \hat{C} \rangle| \leq 2\sqrt{2}$. $\square$

Quantum correlations are therefore **nonlocal**: no local realistic model reproduces them. They are also limited — the no-signalling principle alone would allow $|S| = 4$ (“PR boxes”).

6.4. Instantaneous collapse without signalling

Alice’s measurement collapses Bob’s conditional state instantaneously, whatever the distance. Does this allow faster-than-light communication? No: as shown in Chapter 5, Bob’s reduced state — and hence every statistic he can observe — is independent of Alice’s choice of measurement. The nonlocal correlations become visible only when Alice and Bob compare their records through a classical (light-speed limited) channel. Nonlocality and relativity coexist peacefully.

Note

Experimental Bell tests began with Freedman–Clauser (1972) and Aspect (1982). Two loopholes long remained open: *locality* (settings chosen and outcomes recorded within light-cone separation) and *detection* (losing pairs can fake violations). Both were closed simultaneously in 2015 (Delft, Vienna, NIST). The 2022 Nobel prize in Physics went to Aspect, Clauser and Zeilinger. A CHSH violation is also a resource: it certifies entanglement and randomness without trusting the devices (device-independent QKD and QRNG, Chapter 10).

6.5. Exercises

Exercise 6.1

Show that for the Bell state $|\Phi^+\rangle$, $E(\vec{a}, \vec{b}) = a_x b_x - a_y b_y + a_z b_z$, and find settings in the xz -plane achieving $|S| = 2\sqrt{2}$.

Solution

$\langle \hat{\sigma}_j \otimes \hat{\sigma}_k \rangle_{\Phi^+}$ is 1 for xx and zz , -1 for yy , 0 otherwise. In the xz plane $E = \vec{a} \cdot \vec{b} = \cos(\theta_a - \theta_b)$. Take $\theta_a = 0$, $\theta_{a'} = \pi/2$, $\theta_b = \pi/4$, $\theta_{b'} = -\pi/4$: $S = \cos \frac{\pi}{4} + \cos \frac{\pi}{4} + \cos \frac{\pi}{4} - \cos \frac{3\pi}{4} = 2\sqrt{2}$.

Exercise 6.2

Construct an explicit LHV model reproducing the singlet correlations when both parties measure only along z .

Solution

Let $\lambda = \pm 1$ with probability $\frac{1}{2}$, $A(z, \lambda) = \lambda$, $B(z, \lambda) = -\lambda$. Each outcome is uniformly random and $E(z, z) = -1$. The model fails as soon as several settings are used, as CHSH proves.

6.6. Sources

Prof. Vallone, *Quantum Information and Computing*, lectures 5–7 (the entangled cakes, EPR, local hidden variables, CHSH, instantaneous collapse and no signalling) and the slides [MagicCake_Vallone-expanded](#). Nielsen & Chuang, Section 2.6. Benenti, Casati & Strini, Vol. I, Chapter 2. Original references: A. Einstein,

6. *Nonlocality and Bell inequalities*

B. Podolsky and N. Rosen, Phys. Rev. **47**, 777 (1935); J. S. Bell, Physics **1**, 195 (1964); J. Clauser, M. Horne, A. Shimony and R. Holt, Phys. Rev. Lett. **23**, 880 (1969); P. Kwiat and L. Hardy, Am. J. Phys. **68**, 33 (2000).

7. Detecting entanglement

For pure states, entanglement is easy to diagnose: look at the reduced state. Real sources produce mixed states, and there the question “is this state entangled?” becomes subtle — in general it is even computationally hard. This chapter defines separability for mixed states, explains its operational meaning in terms of local operations and classical communication (LOCC), and presents the two standard tools: the **Peres–Horodecki partial-transpose criterion** and **entanglement witnesses**.

7.1. Pure states: the reduced-state criterion

From Chapter 5, a pure state $|\psi\rangle_{AB}$ is entangled iff $\hat{\rho}_A = \text{Tr}_B |\psi\rangle\langle\psi|$ is mixed. In components, with $|\psi\rangle = \sum_{ab} \psi_{ab} |ab\rangle$ and the coefficient matrix $\Psi = [\psi_{ab}]$,

$$\hat{\rho}_A = \text{Tr}_B [|\psi\rangle_{AB} \langle\psi|] = \sum_{k=0}^{d_B-1} {}_B\langle k|\psi\rangle_{AB} \langle\psi|k\rangle_B = \Psi\Psi^\dagger.$$

Example 7.1 (*The cake state is entangled*)

For $|\psi\rangle = \frac{1}{2} |11\rangle - \sqrt{3/8}(|01\rangle + |10\rangle)$ of Chapter 6, $\Psi = \begin{pmatrix} 0 & -\sqrt{3/8} \\ -\sqrt{3/8} & \frac{1}{2} \end{pmatrix}$ and

$$\hat{\rho}_A = \Psi\Psi^\dagger = \begin{pmatrix} 3/8 & -\frac{1}{2}\sqrt{3/8} \\ -\frac{1}{2}\sqrt{3/8} & 5/8 \end{pmatrix}, \quad \text{Tr } \hat{\rho}_A^2 = \frac{23}{32} < 1.$$

The eigenvalues are $\frac{1}{2}(1 \pm \sqrt{7}/4) \approx 0.83, 0.17$: the state is entangled, but not maximally.

7.2. Mixed states: separability

Definition 7.1 (Separable and entangled mixed states). A state $\hat{\rho}_{AB}$ is **separable** if it can be written as a convex combination of product states,

$$\hat{\rho}_{AB} = \sum_i p_i \hat{\rho}_A^{(i)} \otimes \hat{\rho}_B^{(i)}, \quad p_i \geq 0, \quad \sum_i p_i = 1;$$

otherwise it is **entangled**.

Note that $\hat{\rho}_A$ being mixed no longer implies entanglement: $\hat{1}/4$ is separable but has maximally mixed marginals.

Operational meaning. Separable states are exactly those that Alice and Bob can prepare with **local operations and classical communication** (LOCC): a shared random number i (drawn with probability p_i) tells each to prepare $\hat{\rho}_A^{(i)}$ and $\hat{\rho}_B^{(i)}$ locally. Conversely, LOCC cannot create entanglement from separable states, and cannot increase it on average. Entanglement is therefore the resource that LOCC cannot produce — the currency of teleportation, QKD with entanglement and dense coding.

7. Detecting entanglement

Deciding separability is hard in general (NP-hard for large dimensions), because a state admits infinitely many decompositions. We need criteria.

7.3. Transpose and partial transpose

The **transpose** $\hat{\rho}^T$ (in a fixed basis, $\langle j|\hat{\rho}^T|k\rangle = \langle k|\hat{\rho}|j\rangle$) of a density operator is again a density operator: same eigenvalues. It is a **positive** map. The **partial transpose** acts on one subsystem only:

$$(|a\rangle\langle a'| \otimes |b\rangle\langle b'|)^{T_B} = |a\rangle\langle a'| \otimes |b'\rangle\langle b|, \quad \langle a, b|\hat{\rho}^{T_B}|a', b'\rangle = \langle a, b'|\hat{\rho}|a', b\rangle.$$

Theorem 7.1 (Peres criterion (PPT)). *If $\hat{\rho}_{AB}$ is separable, then $\hat{\rho}^{T_B}$ is positive semidefinite. Equivalently, a negative eigenvalue of $\hat{\rho}^{T_B}$ proves entanglement.*

Proof. $\hat{\rho}^{T_B} = \sum_i p_i \hat{\rho}_A^{(i)} \otimes (\hat{\rho}_B^{(i)})^T$, and each $(\hat{\rho}_B^{(i)})^T$ is a density operator; so $\hat{\rho}^{T_B}$ is itself a separable state, hence positive. $\square$

Theorem 7.2 (Horodecki, 1996). *For 2×2 and 2×3 systems, positivity of the partial transpose is also sufficient for separability. In higher dimensions there exist entangled PPT states (“bound entangled” states, from which no pure entanglement can be distilled).*

The partial transpose is the simplest example of a map that is **positive but not completely positive**: applied to one half of an entangled state, it can produce something that is not a state. This idea returns in Chapter 8.

Example 7.2 (Werner states)

$$\hat{\rho}_W = p |\Psi^-\rangle\langle\Psi^-| + (1-p)\frac{\hat{1}}{4}, \quad 0 \leq p \leq 1.$$

In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, $|\Psi^-\rangle\langle\Psi^-|$ has entries $\frac{1}{2}$ at $(01, 01)$ and $(10, 10)$ and $-\frac{1}{2}$ at $(01, 10)$ and $(10, 01)$. The partial transpose moves the off-diagonal entries to $(00, 11)$ and $(11, 00)$, giving

$$\hat{\rho}_W^{T_B} = \begin{pmatrix} \frac{1-p}{4} & 0 & 0 & -\frac{p}{2} \\ 0 & \frac{1+p}{4} & 0 & 0 \\ 0 & 0 & \frac{1+p}{4} & 0 \\ -\frac{p}{2} & 0 & 0 & \frac{1-p}{4} \end{pmatrix},$$

with eigenvalues $\frac{1+p}{4}$ (three times) and $\frac{1-3p}{4}$. The Werner state is entangled **iff** $p > \frac{1}{3}$ (PPT is necessary and sufficient for two qubits). Note that it violates CHSH only for $p > 1/\sqrt{2}$: there are entangled states with local correlations.

7.4. Entanglement witnesses

The set of separable states is **convex** and closed. By the Hahn–Banach separation theorem, every entangled state can be separated from it by a hyperplane.

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Definition 7.2 (Entanglement witness). A Hermitian operator $\hat{W}$ is an **entanglement witness** if

$$\text{Tr}[\hat{W}\hat{\sigma}_{AB}] \geq 0 \text{ for every separable } \hat{\sigma}_{AB}, \quad \text{Tr}[\hat{W}\hat{\rho}_{AB}] < 0 \text{ for some entangled } \hat{\rho}_{AB}.$$

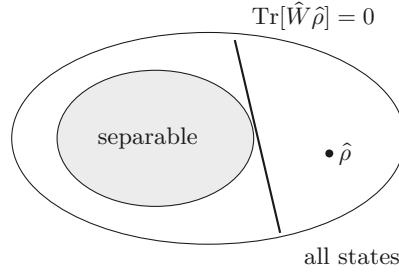


Figure: a witness is a hyperplane leaving all separable states on one side; states on the other side ($\text{Tr}[\hat{W}\hat{\rho}] < 0$) are certified entangled.

A witness is an *observable*: measuring it requires only local measurements (decompose $\hat{W}$ into products of local operators), no full tomography. A negative expectation value proves entanglement; a non-negative one proves nothing. Every entangled state has some witness detecting it, but each witness detects only part of the entangled set.

Example 7.3 (A witness for $|\Phi^+\rangle$)

Let $\hat{W} = \hat{1} \otimes \hat{1} - \hat{\sigma}_x \otimes \hat{\sigma}_x + \hat{\sigma}_y \otimes \hat{\sigma}_y - \hat{\sigma}_z \otimes \hat{\sigma}_z$. Since $|\Phi^+\rangle \langle \Phi^+| = \frac{1}{4}(\hat{1} + \hat{\sigma}_x \hat{\sigma}_x - \hat{\sigma}_y \hat{\sigma}_y + \hat{\sigma}_z \hat{\sigma}_z)$, we have $\hat{W} = 2\hat{1} - 4|\Phi^+\rangle \langle \Phi^+|$. A product state has overlap at most $\frac{1}{2}$ with $|\Phi^+\rangle$, so $\langle \hat{W} \rangle \geq 2 - 4 \cdot \frac{1}{2} = 0$ on product states and, by convexity, on all separable states. On $|\Phi^+\rangle$, $\langle \hat{W} \rangle = -2$.

7.5. Exercises

Exercise 7.1

Show that a negative expectation of a witness on any state $\hat{\rho}'$ implies that $\hat{\rho}'$ is entangled.

Solution

By definition $\text{Tr}[\hat{W}\hat{\sigma}] \geq 0$ for all separable $\hat{\sigma}$. If $\text{Tr}[\hat{W}\hat{\rho}'] < 0$, then $\hat{\rho}'$ cannot be separable.

Exercise 7.2

For $\hat{\rho}_w = (1-p)|\Phi^+\rangle \langle \Phi^+| + \frac{p}{2}(|00\rangle \langle 00| + |11\rangle \langle 11|)$, find for which p the witness above detects entanglement, and compare with the Peres criterion.

Solution

$\langle \Phi^+ | \hat{\rho}_w | \Phi^+ \rangle = (1-p) + \frac{p}{2}(\frac{1}{2} + \frac{1}{2}) = 1 - \frac{p}{2}$, so $\langle \hat{W} \rangle = 2 - 4(1 - \frac{p}{2}) = 2p - 2 < 0$ for all $p < 1$. Partial transpose: $\hat{\rho}_w$ has diagonal $(\frac{1}{2}, 0, 0, \frac{1}{2})$ and coherence $\frac{1-p}{2}$ between $|00\rangle$ and $|11\rangle$; transposition moves it into the $\{|01\rangle, |10\rangle\}$ block $\begin{pmatrix} 0 & \frac{1-p}{2} \\ \frac{1-p}{2} & 0 \end{pmatrix}$, with eigenvalue $-\frac{1-p}{2} < 0$ for $p < 1$. Both criteria agree: entangled iff $p < 1$; at $p = 1$ the state is the separable classical mixture of $|00\rangle$ and $|11\rangle$.

7.6. Sources

Prof. Vallone, *Quantum Information and Computing*, lecture 8 (partial-trace criterion with the cake state, mixed-state entanglement, LOCC, partial transpose, Peres criterion with Werner states, entanglement witnesses) and the evaluated homework (exercise 3). Nielsen & Chuang, Section 2.4.3 and Chapter 12.5. Further reading: R. Horodecki, P. Horodecki, M. Horodecki and K. Horodecki, *Quantum entanglement*, Rev. Mod. Phys. **81**, 865 (2009); A. Peres, Phys. Rev. Lett. **77**, 1413 (1996).

8. Quantum operations and generalized measurements

The postulates of Chapter 4 describe isolated systems. Real qubits interact with an environment: they decohere, lose energy, get entangled with ancillas. The same mathematics describes a system that is only *part* of a larger system evolving unitarily. This chapter derives the **Kraus representation** of the most general physical evolution (a quantum channel, or superoperator), characterises it axiomatically, and does the same for measurements: **generalized measurements** and **POVMs**, which can outperform projective measurements, for example in distinguishing non-orthogonal states.

8.1. From unitaries on a larger system to superoperators

Let the system A start in $|\psi\rangle_A$ and an environment (or ancilla) B in a fixed state $|0\rangle_B$, and let them interact through a unitary $\hat{U}_{AB}$. Expanding the output in a basis $\{|k\rangle_B\}$ of B ,

$$|\Phi\rangle_{AB} = \hat{U}_{AB} |\psi\rangle_A |0\rangle_B = \sum_{k=0}^{d_B-1} (\hat{E}_k |\psi\rangle_A) |k\rangle_B, \quad \hat{E}_k = {}_B\langle k | \hat{U}_{AB} | 0 \rangle_B.$$

The state of A alone is

$$\hat{\rho}'_A = \text{Tr}_B[|\Phi\rangle_{AB}\langle\Phi|] = \sum_k \hat{E}_k \hat{\rho}_A \hat{E}_k^\dagger, \quad \sum_k \hat{E}_k^\dagger \hat{E}_k = \hat{1}_A,$$

where the normalisation follows from unitarity: $\sum_k \hat{E}_k^\dagger \hat{E}_k = \langle 0 | \hat{U}^\dagger \hat{U} | 0 \rangle_B = \hat{1}$. By linearity the formula holds for mixed $\hat{\rho}_A$ too. The operators $\hat{E}_k$ are the **Kraus operators**.

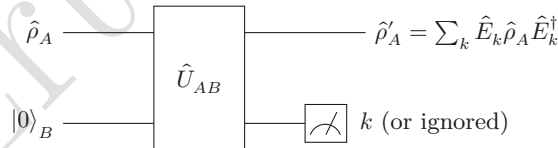


Figure: system and ancilla interact unitarily. Ignoring the ancilla gives a quantum channel on A ; measuring it gives a generalized measurement.

8.2. Superoperators

Definition 8.1 (Superoperator (quantum channel)). A **superoperator** is a map $\chi : \hat{\rho} \mapsto \hat{\rho}' = \chi(\hat{\rho})$ on density operators that is

1. **linear:** $\chi(p_1 \hat{\rho}_1 + p_2 \hat{\rho}_2) = p_1 \chi(\hat{\rho}_1) + p_2 \chi(\hat{\rho}_2)$ (it respects mixtures);
2. **Hermiticity preserving:** $\chi(\hat{\rho})^\dagger = \chi(\hat{\rho})$;
3. **trace preserving:** $\text{Tr} \chi(\hat{\rho}) = 1$;
4. **completely positive:** $\hat{1}_R \otimes \chi$ maps positive operators to positive operators for every reference system R .

Complete positivity is stronger than positivity, and physically necessary: the system may be entangled with a distant partner R untouched by the evolution, and the joint state must remain a state. The transpose is positive but not completely positive — applied to half of $|\Phi^+\rangle$ it produces an operator with a negative eigenvalue (the partial transpose of Chapter 7). It is not a physical evolution.

Theorem 8.1 (Kraus representation). *A map χ satisfies 1–4 if and only if it can be written as*

$$\chi(\hat{\rho}) = \sum_k \hat{E}_k \hat{\rho} \hat{E}_k^\dagger, \quad \sum_k \hat{E}_k^\dagger \hat{E}_k = \hat{\mathbb{1}},$$

with at most d^2 Kraus operators. Every such map arises from a unitary on the system plus an ancilla (Stinespring dilation).

The Kraus operators are not unique: $\{\hat{E}_k\}$ and $\{\hat{F}_j = \sum_k u_{jk} \hat{E}_k\}$, with u unitary, describe the same channel — a different basis for the environment. A unitary evolution is the special case of a single Kraus operator.

Example 8.1 (Depolarizing channel)

With probability $1 - p$ the qubit is untouched; with probability p it is replaced by the maximally mixed state:

$$\chi(\hat{\rho}) = (1 - p)\hat{\rho} + p \frac{\hat{\mathbb{1}}}{2} = \left(1 - \frac{3p}{4}\right)\hat{\rho} + \frac{p}{4}(\hat{\sigma}_x \hat{\rho} \hat{\sigma}_x + \hat{\sigma}_y \hat{\rho} \hat{\sigma}_y + \hat{\sigma}_z \hat{\rho} \hat{\sigma}_z),$$

using $\frac{\hat{\mathbb{1}}}{2} = \frac{1}{4}(\hat{\rho} + \sum_j \hat{\sigma}_j \hat{\rho} \hat{\sigma}_j)$. Kraus operators: $\sqrt{1 - 3p/4} \hat{\mathbb{1}}$ and $\frac{\sqrt{p}}{2} \hat{\sigma}_j$. On the Bloch ball, $\vec{r} \mapsto (1 - p)\vec{r}$: a uniform contraction towards the centre.

Other standard qubit channels: **dephasing** ($\hat{E}_0 = \sqrt{1 - p} \hat{\mathbb{1}}$, $\hat{E}_1 = \sqrt{p} \hat{\sigma}_z$; shrinks r_x, r_y , keeps r_z — loss of coherence, T_2) and **amplitude damping** ($\hat{E}_0 = |0\rangle\langle 0| + \sqrt{1 - \gamma} |1\rangle\langle 1|$, $\hat{E}_1 = \sqrt{\gamma} |0\rangle\langle 1|$; relaxation of $|1\rangle$ to $|0\rangle$, T_1).

8.3. Generalized measurements

Measuring the ancilla instead of discarding it turns the same construction into a measurement on A . If after $\hat{U}_{AB} |\psi\rangle_A |0\rangle_B = \sum_k \hat{M}_k |\psi\rangle_A |k\rangle_B$ we measure B with the projectors $\hat{P}_k = \hat{\mathbb{1}}_A \otimes |k\rangle\langle k|_B$, outcome k occurs with probability

$$p_k = \langle \psi | \hat{M}_k^\dagger \hat{M}_k | \psi \rangle,$$

and A is left in $\hat{M}_k |\psi\rangle / \sqrt{p_k}$.

Definition 8.2 (Generalized measurement). A **generalized measurement** is a set of operators $\{\hat{M}_k\}$ (not necessarily Hermitian, not necessarily projectors) with

$$\sum_k \hat{M}_k^\dagger \hat{M}_k = \hat{\mathbb{1}}.$$

Outcome k occurs with probability $p_k = \text{Tr}[\hat{M}_k^\dagger \hat{M}_k \hat{\rho}]$, and the post-measurement state is $\hat{\rho}'_k = \hat{M}_k \hat{\rho} \hat{M}_k^\dagger / p_k$.

Projective measurements are the special case $\hat{M}_k = \hat{\Pi}_k$. Conversely, the derivation shows that **projective measurements plus unitaries are equivalent to generalized measurements, provided ancillary systems are added** (Naimark's theorem).

Definition 8.3 (POVM). If only the outcome statistics matter, not the post-measurement state, a measurement is described by its **POVM** (positive operator-valued measure) elements

$$\hat{F}_k = \hat{M}_k^\dagger \hat{M}_k, \quad \hat{F}_k \geq 0, \quad \sum_k \hat{F}_k = \hat{1}, \quad p_k = \text{Tr}[\hat{F}_k \hat{\rho}].$$

Any set of positive operators summing to the identity is a valid POVM (take $\hat{M}_k = \sqrt{\hat{F}_k}$). Unlike projectors, POVM elements need not be orthogonal, and there can be more outcomes than the dimension of the space.

Example 8.2 (*Unambiguous discrimination of non-orthogonal states*)

A source emits $|0\rangle$ or $|+\rangle$, which are not orthogonal: no measurement can identify them with certainty every time. A projective measurement always risks misidentification. The POVM

$$\hat{F}_+ = c |1\rangle \langle 1|, \quad \hat{F}_0 = c |-\rangle \langle -|, \quad \hat{F}_? = \hat{1} - \hat{F}_+ - \hat{F}_0, \quad c = \frac{1}{1 + |\langle 0|+\rangle|} = 2 - \sqrt{2},$$

never errs: outcome “+” can only occur for $|+\rangle$ (since $\langle 1|0\rangle = 0$), outcome “0” only for $|0\rangle$; the price is an inconclusive outcome “?”. The success probability $1 - |\langle 0|+\rangle| = 1 - \frac{1}{\sqrt{2}} \approx 0.29$ is optimal (Ivanovic–Dieks–Peres). This trick underlies the B92 and SARG04 QKD protocols (Chapter 10).

8.4. Postulates for interacting systems

1. **States** are density operators $\hat{\rho}$.
2. **Evolution** is described by superoperators, $\chi(\hat{\rho}) = \sum_k \hat{E}_k \hat{\rho} \hat{E}_k^\dagger$ with $\sum_k \hat{E}_k^\dagger \hat{E}_k = \hat{1}$.
3. **Measurements** are sets $\{\hat{M}_k\}$ with $\sum_k \hat{M}_k^\dagger \hat{M}_k = \hat{1}$; outcome k has probability $p_k = \text{Tr}[\hat{M}_k^\dagger \hat{M}_k \hat{\rho}]$.
4. **Collapse**: after outcome k , $\hat{\rho} \mapsto \hat{M}_k \hat{\rho} \hat{M}_k^\dagger / p_k$.

These are not new axioms: they follow from the isolated-system postulates applied to the system plus its environment.

8.5. Exercises

Exercise 8.1

A qubit $|\psi\rangle_A = \cos \frac{\theta}{2} |0\rangle + i \sin \frac{\theta}{2} |1\rangle$ interacts with a qutrit ancilla in $|0\rangle_B$ via $\hat{U} |0\rangle |0\rangle = \frac{1}{2} |0\rangle (|0\rangle + |2\rangle) + \frac{1}{\sqrt{2}} |1\rangle |2\rangle$, $\hat{U} |1\rangle |0\rangle = \frac{1}{2} |0\rangle (|0\rangle - |2\rangle) + \frac{1}{\sqrt{2}} |1\rangle |1\rangle$. (a) Find the Kraus operators. (b) The ancilla is measured in $\{|0\rangle, |1\rangle, |2\rangle\}$: find the outcome probabilities and the post-measurement state for outcome 2.

Solution

(a) $\hat{E}_k = \langle k | \hat{U} | 0 \rangle_B$: $\hat{E}_0 = \frac{1}{2} |0\rangle (\langle 0| + \langle 1|)$, $\hat{E}_1 = \frac{1}{\sqrt{2}} |1\rangle \langle 1|$, $\hat{E}_2 = \frac{1}{2} |0\rangle (\langle 0| - \langle 1|) + \frac{1}{\sqrt{2}} |1\rangle \langle 0|$. One checks $\sum_k \hat{E}_k^\dagger \hat{E}_k = \hat{1}$. (b) With $c = \cos \frac{\theta}{2}$, $s = \sin \frac{\theta}{2}$: $\hat{E}_0 |\psi\rangle = \frac{1}{2}(c + is) |0\rangle$, so $p_0 = \frac{1}{4}$; $\hat{E}_1 |\psi\rangle = \frac{is}{\sqrt{2}} |1\rangle$, so $p_1 = \frac{s^2}{2}$; $\hat{E}_2 |\psi\rangle = \frac{1}{2}(c - is) |0\rangle + \frac{c}{\sqrt{2}} |1\rangle$, so $p_2 = \frac{1}{4} + \frac{c^2}{2}$. The post-measurement state for outcome 2 is $[\frac{1}{2}(c - is) |0\rangle + \frac{c}{\sqrt{2}} |1\rangle] / \sqrt{p_2}$.

Exercise 8.2

Show that the depolarizing channel contracts every Bloch vector by $1 - p$, and that the dephasing channel maps $(r_x, r_y, r_z) \mapsto ((1 - 2p)r_x, (1 - 2p)r_y, r_z)$.

Solution

Depolarizing: $(1 - p)\frac{1}{2}(\hat{1} + \vec{r} \cdot \hat{\sigma}) + p\frac{\hat{1}}{2} = \frac{1}{2}(\hat{1} + (1 - p)\vec{r} \cdot \hat{\sigma})$. Dephasing: $\hat{\sigma}_z \hat{\sigma}_{x,y} \hat{\sigma}_z = -\hat{\sigma}_{x,y}$ and $\hat{\sigma}_z \hat{\sigma}_z \hat{\sigma}_z = \hat{\sigma}_z$, so $(1 - p)\hat{\rho} + p\hat{\sigma}_z \hat{\rho} \hat{\sigma}_z$ multiplies r_x, r_y by $1 - 2p$ and leaves r_z .

8.6. Sources

Prof. Vallone, *Quantum Information and Computing*, lectures 10 (superoperators, Kraus representation, depolarizing channel, generalized measurements, postulates for interacting systems) and 11 (POVM, discrimination of non-orthogonal states), and the evaluated homework (exercise 2). Nielsen & Chuang, Sections 2.2.3–2.2.6 and Chapter 8. Benenti, Casati & Strini, Vol. I, Chapter 5. Preskill, *Lecture Notes*, Chapter 3.

Part II.

Part II: Quantum information

9. Classical and quantum Shannon theory

How much information does a quantum system carry? A qubit has a continuum of states, yet intuition suggests it cannot store more than one bit retrievably. This chapter recalls the classical entropies of Shannon — compression rate, mutual information, channel capacity — introduces their quantum counterpart, the **von Neumann entropy**, highlights where quantum and classical behaviour part ways (negative conditional entropy), and proves the operational answer: the **Holevo bound** on accessible information.

9.1. Classical information theory in brief

Definition 9.1 (Shannon entropy). For a random variable X with distribution p_x ,

$$H(X) = - \sum_x p_x \log_2 p_x.$$

$H(X)$ is the average surprise of an outcome and, operationally, the **compression rate**: by Shannon's **noiseless coding theorem**, n i.i.d. copies of X can be compressed into about $nH(X)$ bits and no fewer, for $n \rightarrow \infty$. It ranges from 0 (deterministic) to $\log_2 |X|$ (uniform). For a binary variable with $p(1) = Q$ it is the **binary entropy** $h_2(Q) = -Q \log_2 Q - (1-Q) \log_2 (1-Q)$.

For two variables:

- **joint entropy** $H(X, Y) = - \sum_{x,y} p_{xy} \log_2 p_{xy}$;
- **conditional entropy** $H(X|Y) = H(X, Y) - H(Y)$: the additional bits needed to specify X when Y is known — the residual ignorance about X ;
- **mutual information** $I(X : Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X, Y)$: how much learning Y reduces the bits needed to specify X .

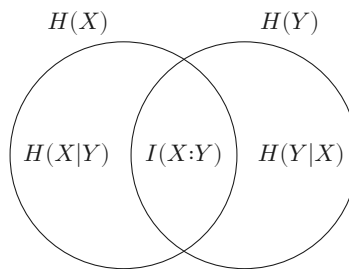


Figure: entropies of two variables. The union is $H(X, Y)$; classically every region is non-negative.

Theorem 9.1 (Noisy channel coding theorem). A channel defined by conditional probabilities $p(y|x)$ has **capacity**

$$C = \max_{p_x} I(X : Y).$$

Every rate $R < C$ can be transmitted with vanishing error probability, and no rate $R > C$ can.

For the binary symmetric channel with flip probability Q , $C = 1 - h_2(Q)$, achieved by uniform inputs. This formula reappears in QKD as the information Alice and Bob share through a channel with error rate Q .

9.2. Von Neumann entropy

Definition 9.2 (Von Neumann entropy).

$$S(\hat{\rho}) = -\text{Tr}[\hat{\rho} \log_2 \hat{\rho}] = -\sum_j \lambda_j \log_2 \lambda_j,$$

where λ_j are the eigenvalues of $\hat{\rho}$: the Shannon entropy of its spectrum.

Proposition 9.2 (Properties). 1. $S(\hat{\rho}) = 0$ iff $\hat{\rho}$ is pure.

2. **Unitary invariance:** $S(\hat{U}\hat{\rho}\hat{U}^\dagger) = S(\hat{\rho})$.

3. $0 \leq S(\hat{\rho}) \leq \log_2 d$, with the maximum for $\hat{1}/d$.

4. **Concavity:** $S(\sum_k p_k \hat{\rho}_k) \geq \sum_k p_k S(\hat{\rho}_k)$ — mixing increases ignorance.

5. **Subadditivity:** $S(AB) \leq S(A) + S(B)$, with equality iff $\hat{\rho}_{AB} = \hat{\rho}_A \otimes \hat{\rho}_B$ (as for Shannon entropy).

6. **Araki–Lieb (triangle) inequality:** $S(AB) \geq |S(A) - S(B)|$.

7. For a **pure bipartite state**, $S(A) = S(B)$ (both reduced states have the Schmidt coefficients as spectrum).

For a qubit with Bloch vector $\vec{r}$, $S = h_2(\frac{1+|\vec{r}|}{2})$.

Point 6 departs from the classical relation $H(X, Y) \geq \max(H(X), H(Y))$. For a Bell state, $S(AB) = 0$ while $S(A) = S(B) = 1$: the whole is less uncertain than its parts. Consequently the **quantum conditional entropy**

$$S(A|B) = S(AB) - S(B)$$

can be **negative**: $S(A|B) = -1$ for a Bell state. Negative conditional entropy is a signature of entanglement (it is impossible for separable states) and has an operational meaning in quantum state merging. For pure states, $S(A) = S(B)$ is the standard measure of entanglement, the **entanglement entropy**.

Definition 9.3 (Quantum mutual information). $I(A : B) = S(A) + S(B) - S(AB) \geq 0$. For a Bell state $I(A : B) = 2$: twice the classical maximum for two bits.

9.3. Accessible information and the Holevo bound

Alice encodes a classical letter x (probability p_x) into a state $\hat{\rho}_x$ and sends it to Bob, who performs a POVM $\{\hat{F}_y\}$. The outcome statistics are $p(y|x) = \text{Tr}[\hat{F}_y \hat{\rho}_x]$ (after the channel, if any). The **accessible information** is the best mutual information Bob can achieve:

$$I_{acc} = \max_{\{\hat{F}_y\}} I(X : Y) = \max_{\{\hat{F}_y\}} [H(Y) - H(Y|X)].$$

Theorem 9.3 (Holevo bound (1973)). For the ensemble $\{p_x, \hat{\rho}_x\}$ with average state $\hat{\rho} = \sum_x p_x \hat{\rho}_x$,

$$I_{acc} \leq \chi \equiv S(\hat{\rho}) - \sum_x p_x S(\hat{\rho}_x).$$

Since $\chi \leq S(\hat{\rho}) \leq \log_2 d$, by sending n qubits Alice can convey at most n bits of classical information — despite the continuum of states a qubit can occupy. For pure states $\chi = S(\hat{\rho})$, and the bound coincides with Schumacher’s quantum compression rate. The bound is generally not attained by measuring letter by letter, but it is achievable asymptotically with block coding (Holevo–Schumacher–Westmoreland).

Example 9.1 (*Two non-orthogonal pure states*)

Alice sends $|\theta_{0,1}\rangle = \cos \frac{\theta}{2} |0\rangle \pm \sin \frac{\theta}{2} |1\rangle$ with equal probabilities. The average state is diagonal, $\hat{\rho} = \cos^2 \frac{\theta}{2} |0\rangle\langle 0| + \sin^2 \frac{\theta}{2} |1\rangle\langle 1|$, so

$$\chi = S(\hat{\rho}) = h_2\left(\cos^2 \frac{\theta}{2}\right) \leq 1.$$

Bob measures with a projective measurement $\hat{\Pi}_{\pm} = \frac{1}{2}(\hat{1} \pm \vec{n} \cdot \hat{\sigma})$; by symmetry the optimum is $\vec{n} = \hat{x}$, for which $p(\pm|\theta_0) = \frac{1}{2}(1 \pm \sin \theta)$: a binary symmetric channel with error $\frac{1}{2}(1 - \sin \theta)$, so

$$I(X : Y) = 1 - h_2\left(\frac{1 - \sin \theta}{2}\right) \leq h_2\left(\frac{1 + \cos \theta}{2}\right) = \chi,$$

with equality only for orthogonal states ($\theta = \pi/2$), where both equal 1.

Example 9.2 (*The trine*)

Alice sends $|\phi_0\rangle = |0\rangle$, $|\phi_{1,2}\rangle = \frac{1}{2}|0\rangle \pm \frac{\sqrt{3}}{2}|1\rangle$ with probability $\frac{1}{3}$ each — three states at 120° on a great circle of the Bloch sphere. Since $\hat{\rho} = \hat{1}/2$, the Holevo quantity is $\chi = 1$. The optimal measurement is the “anti-trine” POVM

$$\hat{F}_y = \frac{2}{3} |\phi_y^\perp\rangle\langle \phi_y^\perp|,$$

with $|\phi_0^\perp\rangle = |1\rangle$, $|\phi_{1,2}^\perp\rangle = \frac{\sqrt{3}}{2}|0\rangle \mp \frac{1}{2}|1\rangle$. Outcome y *excludes* state y : $p(y|x) = 0$ for $y = x$ and $\frac{1}{2}$ otherwise. Then $H(X) = H(Y) = \log_2 3$ and $H(X, Y) = \log_2 3 + 1$, giving

$$I_{acc} = \log_2 3 - 1 \approx 0.585 < \chi = 1.$$

A three-outcome POVM on a qubit beats any two-outcome projective measurement here — a concrete advantage of generalized measurements.

9.4. Exercises

Exercise 9.1

Compute $S(A)$, $S(B)$, $S(AB)$, $S(A|B)$ and $I(A : B)$ for the Werner state $\hat{\rho}_W = p |\Psi^-\rangle\langle \Psi^-| + (1-p)\hat{1}/4$ and find when $S(A|B) < 0$.

Solution

Marginals are $\hat{1}/2$: $S(A) = S(B) = 1$. Eigenvalues of $\hat{\rho}_W$: $\frac{1+3p}{4}$ once and $\frac{1-p}{4}$ three times, so $S(AB) = -\frac{1+3p}{4} \log_2 \frac{1+3p}{4} - \frac{3(1-p)}{4} \log_2 \frac{1-p}{4}$. Then $S(A|B) = S(AB) - 1$ and $I(A : B) = 2 - S(AB)$. $S(A|B) < 0$ iff $S(AB) < 1$, which happens for $p \gtrsim 0.75$ — a stricter condition than entanglement ($p > \frac{1}{3}$): negative conditional entropy implies entanglement, not vice versa.

Exercise 9.2

Show that for the BB84 states $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ sent with equal probability, $\chi = 1$, and that Bob measuring in the z basis obtains $I(X : Y) = \frac{1}{2}$ bit about the four-valued letter.

Solution

$\hat{\rho} = \mathbb{1}/2$ and all states are pure: $\chi = 1$. Measuring z : for $x \in \{0, 1\}$ Bob learns x ; for $x \in \{+, -\}$ he gets a uniformly random bit. $H(X) = 2$, $H(X|Y) = 1.5$ (given y , x is y with probability $\frac{1}{2}$ or one of $\pm$ with probability $\frac{1}{4}$ each), so $I = 0.5$ bit — far below χ , and still below the 1 bit Bob could get if the basis were announced.

9.5. Sources

Prof. Vallone, *Quantum Information and Computing*, lecture 11 (POVM, classical information theory recap, von Neumann entropy and its properties, quantum mutual information, accessible information and the Holevo bound, the examples with two and three non-orthogonal states) and exercise sheet **exercise1** (accessible information of the trine). Nielsen & Chuang, Chapters 11–12. Preskill, *Lecture Notes*, Chapter 10. Benenti, Casati & Strini, Vol. I, Chapter 5. Original reference: A. Holevo, *Probl. Peredachi Inf.* **9**, 3 (1973).

10. No-cloning and quantum key distribution

Quantum mechanics forbids copying an unknown state and guarantees that measuring disturbs. Turned around, these limitations become a security guarantee: any eavesdropper on a quantum channel leaves traces. **Quantum key distribution** (QKD) exploits this to let two parties grow a shared secret key whose security rests on physics, not on computational assumptions. This chapter proves the no-cloning theorem, describes BB84 and its descendants, computes secret-key rates, discusses real implementations (lossy channels, weak laser pulses, decoy states), and closes with quantum random number generation.

10.1. Why quantum cryptography

Classical public-key cryptography (RSA, Diffie–Hellman) relies on problems believed to be computationally hard. A classical algorithmic breakthrough could break it, and a quantum computer *will* break RSA (Shor’s algorithm, Chapter 13). Post-quantum cryptography replaces the hard problems; QKD replaces the assumption altogether.

Definition 10.1 (One-time pad). To encrypt an n -bit message m , Alice adds bitwise (XOR) an n -bit uniformly random key k : $c = m \oplus k$. Bob decrypts with $m = c \oplus k$. If the key is secret, random, as long as the message and never reused, the ciphertext is statistically independent of the message (Shannon, 1949): perfect secrecy.

The problem is distributing the key. QKD solves exactly this: it lets Alice and Bob **exchange random and secret keys**, which are then used in a one-time pad or another symmetric cipher.

10.2. The no-cloning theorem

Theorem 10.1 (No-cloning (Wootters–Zurek, Dieks, 1982)). *There is no unitary $\hat{U}$ and fixed blank state $|e\rangle$ such that $\hat{U}|\psi\rangle|e\rangle = |\psi\rangle|\psi\rangle$ for every state $|\psi\rangle$.*

Proof. Suppose it worked for $|\psi\rangle$ and $|\phi\rangle$. Taking the scalar product of $\hat{U}|\psi\rangle|e\rangle = |\psi\rangle|\psi\rangle$ with $\hat{U}|\phi\rangle|e\rangle = |\phi\rangle|\phi\rangle$ and using unitarity,

$$\langle\phi|\psi\rangle\langle e|e\rangle = \langle\phi|\psi\rangle^2 \implies \langle\phi|\psi\rangle \in \{0, 1\}.$$

Only identical or orthogonal states can be cloned by the same machine. $\square$

(Allowing ancillas and measurements, i.e. general channels, does not help.) Orthogonal states *can* be copied — that is classical information. The theorem is consistent with no-signalling: a cloner would let Bob distinguish Alice’s choice of basis on half of an entangled pair, allowing superluminal communication.

10.3. The BB84 protocol

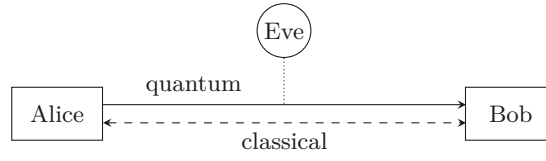


Figure: QKD setting. Eve can do anything to the quantum signals, but can only listen to the authenticated classical channel.

The **quantum channel** is public and not authenticated: Eve can interact with the signals arbitrarily. The **classical channel** is public but authenticated: Eve can read but not modify the messages (authentication needs a short pre-shared key, so QKD is really *key growing*).

- Definition 10.2** (BB84 (Bennett–Brassard, 1984)).
1. For each signal Alice picks a random bit and a random basis, $Z = \{|0\rangle, |1\rangle\}$ or $X = \{|+\rangle, |-\rangle\}$, and sends the corresponding state (e.g. polarisation H/V or D/A).
 2. Bob measures each signal in a randomly chosen basis Z or X .
 3. **Sifting**: over the classical channel they announce the bases (not the bits) and keep only the rounds with matching bases — about half.
 4. **Parameter estimation**: they reveal a random subset of the sifted bits to estimate the **quantum bit error rate** (QBER) Q .
 5. **Error correction** and **privacy amplification** turn the remaining bits into a shorter, identical and secret key.

Example 10.1 (Intercept-resend attack)

Eve measures each photon in a random basis and resends the state she found. With probability $\frac{1}{2}$ she guesses the right basis and causes no error; with probability $\frac{1}{2}$ she uses the wrong basis, resends a state of the wrong basis, and Bob (in the sifted rounds) gets a random result — an error with probability $\frac{1}{2}$. The QBER becomes $\frac{1}{2} \cdot \frac{1}{2} = 25\%$, and Eve is detected. Any attempt to gain information disturbs the states, because the two bases are non-commuting (mutually unbiased) and the states cannot be cloned.

Efficient BB84. Nothing forces the bases to be equally likely. Alice and Bob use Z with probability $p_z \rightarrow 1$ for key generation and X with small probability $p_x = 1 - p_z$ to detect Eve; the sifting probability p_z^2 approaches 1 instead of $\frac{1}{2}$.

10.4. General structure and security

Prepare-and-measure protocols encode a string $S_n = (s_1, \dots, s_n)$ into $|\Psi(S_n)\rangle_A$, usually $|\psi(s_1)\rangle \otimes \dots \otimes |\psi(s_n)\rangle$; the channel $\mathcal{E}$ (including Eve) delivers $\mathcal{E}(|\Psi(S_n)\rangle)$; Bob decodes with a POVM. Security requires a set of **non-orthogonal states**: orthogonal states could be copied. **Entanglement-based** protocols distribute $|\Phi\rangle = \frac{1}{\sqrt{d}} \sum_{S_n} |S_n\rangle_A \otimes |\Psi(S_n)\rangle_B$; measuring her half, Alice remotely prepares Bob's state. The two pictures are equivalent; in particular no key can be extracted from an entanglement-breaking channel.

After the measurements Alice and Bob hold correlated **raw keys**. Classical post-processing — parameter estimation, sifting (yielding N **sifted** bits, partially correlated and partially secret), error correction, privacy amplification (yielding ℓ secret bits) — defines the **secret fraction**

$$r = \lim_{N \rightarrow \infty} \frac{\ell}{N}.$$

Definition 10.3 (Composable ε -security). A key K is ε -**secure** if the joint state of key and Eve is ε -close in trace distance to an ideal key — uniform and uncorrelated with Eve:

$$\|\hat{\rho}_{KE} - \hat{\tau}_K \otimes \hat{\rho}_E\|_1 \leq \varepsilon.$$

The definition is **composable**: an ε -secure key used in an ε' -secure task gives an $(\varepsilon + \varepsilon')$ -secure procedure, whatever the application.

QKD can achieve **unconditional security**, under explicit conditions: Eve cannot intrude into Alice's and Bob's devices; their random number generators are trusted; the classical channel is authenticated; the devices are trusted and characterised. Attacks on real systems target these assumptions (detector blinding, Trojan horses) — the motivation for device-independent and measurement-device-independent QKD.

10.5. A zoo of protocols

- **E91 / BBM92** (Ekert 1991; Bennett, Brassard, Mermin 1992): the entangled version of BB84. A source distributes singlets (any Bell state works); both parties measure Z or X (with $p_z \rightarrow 1$); key from $Z_A Z_B$ rounds after Bob flips his bits; Eve detected from $X_A X_B$ rounds (Ekert's original version tests a CHSH inequality instead).
- **SARG04** (Scarani, Acín, Ribordy, Gisin): the same four states as BB84 but a different encoding — bit 0 is the Z basis, bit 1 the X basis. Alice announces a pair of non-orthogonal states containing the one she sent; Bob keeps only conclusive results (cf. unambiguous discrimination, Chapter 8). It is more robust against photon-number-splitting attacks.
- **B92** uses just two non-orthogonal states and an unambiguous-discrimination measurement.
- **Six-state protocol**: three mutually unbiased bases X, Y, Z . Channel estimation becomes tomographically complete and Eve's information is more constrained.
- **Qudit protocols**: generalised Pauli operators $\hat{U}_{jk} = \sum_n e^{2\pi i n k / d} |n + j\rangle \langle n|$ define bases in dimension d . The two-basis protocol uses eigenstates of $\hat{U}_{01}$ and $\hat{U}_{10}$; for prime d , eigenstates of $\hat{U}_{01}$ and $\hat{U}_{1k}$ give $d + 1$ **mutually unbiased bases** ($|\langle e_i | f_j \rangle|^2 = 1/d$). Higher dimension means more bits per photon and more noise tolerance.
- **Continuous-variable** protocols encode in field quadratures measured by homodyne detection.
- **Distributed-phase-reference** protocols — **DPS** (differential phase shift: bits in the phase difference $0, \pi$ of consecutive coherent pulses, decoded by an unbalanced interferometer) and **COW** (coherent one-way: bits in the time slot of a pulse, security from monitoring coherence between successive non-empty pulses).
- **Round-robin DPS** (“no-disturbance”): a superposition of L pulses with phases ± 1 ; Bob measures the XOR of two random pulses. Eve's information is bounded *regardless of the disturbance* she causes.

10.6. Secret key rates

In the asymptotic limit and for one-way post-processing (Csiszár–Körner, Devetak–Winter),

$$r = I_{AB} - I_E, \quad I_E = \min(I_{EA}, I_{EB}),$$

the information Alice and Bob share minus what Eve knows (with reverse reconciliation, Eve's information on Bob's string).

Eve's capabilities define the attack model: **individual** attacks (each signal attacked independently, ancillas measured before post-processing: $I_E = \max I(A : E)$); **collective** attacks (ancillas stored in a quantum memory and measured collectively at the end: $I_E = \max \chi(A : E)$, with $\chi(A : E) = S(\hat{\rho}_E) - \sum_a p_a S(\hat{\rho}_{E|a})$);

coherent attacks (no restriction). For the main protocols, the bound for coherent attacks coincides with the collective one (de Finetti arguments).

Proposition 10.2 (BB84 key rate). *With QBER Q , $I_{AB} = 1 - h_2(Q)$; real error correction leaks $\text{leak}_{EC} = f h_2(Q)$ with efficiency $f \geq 1$. Eve can gain information only by introducing errors, at most $I_E = h_2(Q)$. Hence*

$$r_{BB84} = 1 - f h_2(Q) - h_2(Q).$$

With $f = 1$, $r > 0$ iff $h_2(Q) < \frac{1}{2}$, i.e. $Q < 11\%$. For the **six-state protocol** the same I_{AB} holds but Eve knows less:

$$I_E(Q) = Q + (1 - Q) h_2\left(\frac{1 - 3Q/2}{1 - Q}\right),$$

and the threshold rises to about 12.6%. For qudits, with $h_d(Q) = -Q \log_2 \frac{Q}{d-1} - (1-Q) \log_2 (1-Q)$, the two-basis protocol gives $r = \log_2 d - f h_d(Q) - h_d(Q)$, and the $(d+1)$ -basis protocol tolerates even more noise.

10.7. Implementations

Encodings. *Polarisation* ($|H\rangle, |V\rangle$ and diagonals, or circular $|R, L\rangle = |H\rangle \mp i|V\rangle$) is natural in free space. *Time-bin* encoding, $|\psi\rangle = (|S\rangle + e^{i\varphi}|L\rangle)/\sqrt{2}$ with S/L a short and a long path of an unbalanced interferometer, is robust in fibres; the relative phase carries the information. *Orbital angular momentum* (phase $e^{im\phi}$, $m \in \mathbb{Z}$) offers large d for qudit protocols. Self-compensating polarisation encoders (e.g. the “POGNAC” Sagnac design) achieve intrinsic QBER below 0.1%.

Channels. Transmission decays exponentially in fibre, $t_{\text{fiber}} = t_0 10^{-\alpha\ell/10}$ with $\alpha \approx 0.2$ dB/km at 1550 nm, but only quadratically (diffraction) in vacuum, $t_{\text{vacuum}} \propto 1/\ell^2$. Over 1000 km, fibre losses are ~ 200 dB while a satellite link loses 50–80 dB: the motivation for space QKD (the Micius satellite, the European EuroQCI initiative). Atmospheric turbulence makes free-space transmission fluctuate (log-normal statistics), which adaptive selection of high-transmissivity windows can exploit.

Detectors. Single-photon detectors (avalanche photodiodes in Geiger mode, superconducting nanowires) are characterised by **quantum efficiency** η_B (click probability per photon), **dark-count** probability p_d (clicks without photons — noise that dominates at long distance), and **dead time**. These parameters are interdependent.

Weak coherent pulses. True single-photon sources are hard; attenuated lasers are used instead. A coherent state $|\alpha\rangle = e^{-\mu/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$ with mean photon number $\mu = |\alpha|^2$, after phase randomisation, becomes a mixture of Fock states with Poisson weights:

$$\hat{\rho} = \int_0^{2\pi} \frac{d\theta}{2\pi} |e^{i\theta}\sqrt{\mu}\rangle \langle e^{i\theta}\sqrt{\mu}| = \sum_n P_n(\mu) |n\rangle \langle n|, \quad P_n(\mu) = e^{-\mu} \frac{\mu^n}{n!}.$$

Multi-photon pulses enable the **photon-number-splitting** (PNS) attack: Eve measures the photon number, keeps one photon of every multi-photon pulse (learning the bit after basis reconciliation, without errors), and blocks single-photon pulses to keep the overall loss unchanged. The secret key comes only from single-photon events:

$$K = R_0 + R_1[1 - h_2(e_1)] - \text{leak}_{EC}$$

(GLLP), where R_n is the detection rate of n -photon pulses and e_1 the single-photon error rate — quantities not directly measured. Pessimistic estimates make the rate drop quickly with distance.

Decoy states (Hwang; Lo, Ma, Chen; Wang, 2005) fix this. Alice randomly varies the intensity among a signal μ (e.g. 0.5), a decoy ν (e.g. 0.1) and vacuum. Let Y_n be the **yield** of n -photon pulses (probability of a click given n photons were sent) and e_n their error rate. Eve cannot tell which intensity an n -photon pulse

came from, so Y_n and e_n are the same for all intensities, while the measured **gains** and QBERs are Poisson mixtures, $G_x = \sum_n P_n(x) Y_n$ and $E_x G_x = \sum_n P_n(x) e_n Y_n$. The vacuum decoy measures Y_0 , and the other two give (Ma, Qi, Zhao, Lo, 2005)

$$Y_1 \geq Y_1^L = \frac{\mu}{\mu\nu - \nu^2} \left[G_\nu e^\nu - G_\mu e^\mu \frac{\nu^2}{\mu^2} - \frac{\mu^2 - \nu^2}{\mu^2} Y_0 \right], \quad e_1 \leq e_1^U = \frac{E_\nu G_\nu e^\nu - \frac{1}{2} Y_0}{Y_1^L \nu}.$$

The single-photon contribution to the key is then bounded with $R_1 \geq \mu e^{-\mu} Y_1^L$ and $e_1 \leq e_1^U$, and the key rate approaches that of an ideal single-photon source. Decoy-state BB84 is the standard in commercial systems.

10.8. Quantum random number generators

Random numbers are needed everywhere: cryptography (including QKD itself), simulation, gaming. **Pseudo-random** generators are deterministic algorithms — fast and simple, but periodic and possibly correlated; flawed generators (IBM’s RANDU, $V_{k+1} = 65539 V_k \bmod 2^{31}$) and deliberate backdoors (the NIST-certified Dual_EC_DRBG) have caused real damage. Classical physical generators (coins, thermal noise) are unpredictable only because of ignorance of initial conditions. “Anyone who attempts to generate random numbers by deterministic means is, of course, living in a state of sin” (von Neumann).

A **QRNG** uses the intrinsic randomness of quantum measurement: outcomes are unpredictable even with complete knowledge of the state. For a trusted device measuring $|\psi\rangle$ in a basis $\{|z\rangle\}$, the relevant quantity is the **min-entropy**

$$H_\infty(Z) = -\log_2 \max_z P_z, \quad P_z = |\langle z|\psi\rangle|^2,$$

related to the guessing probability $p_{\text{guess}} = 2^{-H_\infty}$; randomness extractors distill about H_∞ uniform bits per measurement.

For security applications the source may be correlated with an adversary, who may even prepare it. The relevant quantity is the conditional min-entropy $H_{\min}(Z|E)$. **Source-device-independent** QRNGs bound it without trusting the source, using an entropic uncertainty relation for mutually unbiased bases in dimension d (Tomamichel–Renner):

$$H_{\min}(Z|E) + H_{\max}(X) \geq \log_2 d.$$

Randomly measuring in the conjugate basis X (only to monitor) certifies the randomness of the Z outcomes: $H_{\min}(Z|E) \geq \log_2 d - 2 \log_2 \sum_x \sqrt{p_x}$. Implementations with heterodyne detection of the vacuum reach tens of Gbit/s of certified randomness. This answers the basic question “how do I distinguish $|\psi\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$ (quantum randomness) from $\hat{\rho} = \frac{1}{2}(|H\rangle\langle H| + |V\rangle\langle V|)$ (classical noise)?” — by measuring in a second basis.

10.9. Exercises

Exercise 10.1

Eve performs intercept-resend on a fraction η of the photons in BB84. What QBER does she cause and how much information does she get on the sifted key? Compare with the condition $r > 0$.

Solution

On attacked photons the QBER is 25%, so $Q = \eta/4$. Eve’s measurement basis is right half the time, so she knows the bit on a fraction $\eta/2$ of the sifted key and nothing on the rest (a wrong-basis measurement is uncorrelated with Alice’s bit). Since $h_2(Q)$ upper-bounds Eve’s information in the

security proof, $r = 1 - 2h_2(\eta/4) > 0$ requires $\eta/4 < 0.11$, i.e. $\eta < 0.44$: beyond that, Alice and Bob abort.

Exercise 10.2

Show that a QKD protocol using only two orthogonal states is insecure.

Solution

Orthogonal states can be perfectly distinguished and copied: Eve measures in the encoding basis, learns every bit, and resends an identical state, causing no errors. Security requires non-orthogonal states, for which any information gain implies disturbance.

10.10. Sources

Prof. Vallone, *Quantum Information and Computing*, lectures 12–14 and the decks `Vallone_QKD_lecture1` (one-time pad, BB84, intercept-resend, classical and quantum channels, post-processing, security), `Vallone_QKD_lecture13` (protocols and key rates), `Vallone_QKD_implementation_lecture14` (encodings, channels, detectors, weak coherent pulses, PNS, decoy states) and `Vallone_QI@DEI_2025.26lecture24` (QRNG, satellite quantum communication). Nielsen & Chuang, Sections 12.1 and 12.6. Further reading: N. Gisin, G. Ribordy, W. Tittel and H. Zbinden, *Quantum cryptography*, Rev. Mod. Phys. **74**, 145 (2002); V. Scarani *et al.*, *The security of practical quantum key distribution*, Rev. Mod. Phys. **81**, 1301 (2009); W. Wootters and W. Zurek, Nature **299**, 802 (1982).

11. Teleportation and entanglement swapping

Entanglement is a resource that can be consumed to perform tasks impossible with local operations and classical communication alone. This chapter presents the three canonical protocols: **teleportation** (sending an unknown qubit using one shared Bell pair and two classical bits), **dense coding** (the reverse trade, two classical bits through one qubit), and **entanglement swapping** (entangling particles that never interacted). All three rest on the **Bell-state measurement**, whose optical implementation is discussed first.

11.1. Bell-state measurement

A **Bell-state measurement** (BSM) is a projective measurement of two qubits in the Bell basis $\{|\Phi^\pm\rangle, |\Psi^\pm\rangle\}$. With gates (Chapter 12) it is easy: a CNOT followed by a Hadamard on the control maps the Bell basis to the computational basis,

$$|\Phi^+\rangle \mapsto |00\rangle, \quad |\Phi^-\rangle \mapsto |10\rangle, \quad |\Psi^+\rangle \mapsto |01\rangle, \quad |\Psi^-\rangle \mapsto |11\rangle,$$

and two single-qubit measurements finish the job.

With linear optics (photons, polarisation qubits) there is no deterministic CNOT, but interference does part of the work. Send the two photons onto the two inputs of a **50:50 beam splitter**. Because photons are bosons, the total state (polarisation $\otimes$ spatial mode) must be symmetric. The singlet $|\Psi^-\rangle$ is the only antisymmetric polarisation state, so its spatial part must be antisymmetric: the photons **always exit from different ports** (antibunching). The three symmetric Bell states give photons exiting from the *same* port (Hong–Ou–Mandel bunching). A coincidence between the two outputs thus heralds $|\Psi^-\rangle$. Adding **polarising beam splitters** at the outputs also identifies $|\Psi^+\rangle$ (both photons in the same port but with orthogonal polarisations), while $|\Phi^\pm\rangle$ remain indistinguishable. With linear optics and no ancillas, a BSM succeeds at most **50%** of the time.

11.2. Teleportation

Alice holds a qubit in an unknown state $|\psi\rangle_C = \alpha|0\rangle + \beta|1\rangle$ and wants to transfer it to Bob. She cannot measure it (one shot gives at most one bit and destroys it) nor clone it. With a shared Bell pair $|\Phi^+\rangle_{AB}$ (Alice holds A , Bob holds B) she can.

Expanding C and A in the Bell basis,

$$|\psi\rangle_C |\Phi^+\rangle_{AB} = \frac{1}{2} \left[|\Phi^+\rangle_{CA} |\psi\rangle_B + |\Phi^-\rangle_{CA} \hat{Z} |\psi\rangle_B + |\Psi^+\rangle_{CA} \hat{X} |\psi\rangle_B + |\Psi^-\rangle_{CA} \hat{X} \hat{Z} |\psi\rangle_B \right],$$

where $\hat{Z} |\psi\rangle = \alpha|0\rangle - \beta|1\rangle$, $\hat{X} |\psi\rangle = \beta|0\rangle + \alpha|1\rangle$, $\hat{X} \hat{Z} |\psi\rangle = -\beta|0\rangle + \alpha|1\rangle$. Nothing has happened yet — this is just a rewriting. Now:

1. Alice performs a BSM on C and A ; each of the four outcomes occurs with probability $\frac{1}{4}$, independently of $|\psi\rangle$.
2. Bob's qubit collapses to $\hat{\sigma}_k |\psi\rangle$, a known Pauli rotation of $|\psi\rangle$.
3. Alice sends the two-bit outcome to Bob over a classical channel.

11. Teleportation and entanglement swapping

4. Bob applies the correction $\hat{1}$, $\hat{Z}$, $\hat{X}$ or $\hat{Z}\hat{X}$ and recovers $|\psi\rangle_B$ exactly.

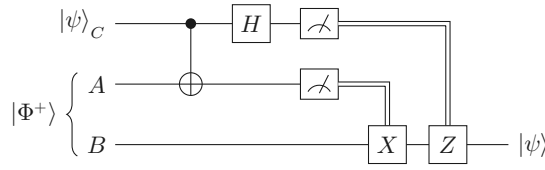


Figure: teleportation circuit. CNOT + Hadamard + two measurements implement the Bell-state measurement; double lines are classical bits controlling Bob's corrections.

Remark (What teleportation does not do).

- **No faster-than-light signalling:** before receiving the two bits, Bob's qubit is maximally mixed ($\frac{1}{4} \sum_k \hat{\sigma}_k |\psi\rangle \langle \psi| \hat{\sigma}_k^\dagger = \hat{1}/2$), independent of $|\psi\rangle$.
- **No cloning:** the original is destroyed by Alice's measurement; only one copy exists at any time.
- **No matter transfer:** the information (the state) moves, not the particle.

The resource count is exact: **1 ebit + 2 classical bits = 1 qubit transmitted**. Teleportation was demonstrated with photons in 1997 (Innsbruck, Rome), later over 143 km in free space and from ground to the Micius satellite. It is the basic primitive of quantum networks and of measurement-based quantum computation.

11.3. Dense coding

The converse trade: with a shared $|\Phi^+\rangle$, Alice can send **two classical bits by transmitting one qubit**. She applies $\hat{1}$, $\hat{Z}$, $\hat{X}$ or $\hat{X}\hat{Z}$ to her half, producing the four orthogonal Bell states

$$|\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle \text{ (up to a sign),}$$

and sends her qubit to Bob, who performs a BSM and reads two bits. This does not violate the Holevo bound (Chapter 9): two qubits are involved in total, and the one sent earlier as part of the entangled pair carries no information by itself. Formally, **1 ebit + 1 qubit = 2 classical bits**.

11.4. Entanglement swapping

Suppose a source prepares $|\Phi^+\rangle_{12}$ and another prepares $|\Phi^+\rangle_{34}$, with photons 1, 4 far apart and 2, 3 sent to a middle station. Regrouping,

$$|\Phi^+\rangle_{12} |\Phi^+\rangle_{34} = \frac{1}{2} \left[|\Phi^+\rangle_{14} |\Phi^+\rangle_{23} + |\Phi^-\rangle_{14} |\Phi^-\rangle_{23} + |\Psi^+\rangle_{14} |\Psi^+\rangle_{23} + |\Psi^-\rangle_{14} |\Psi^-\rangle_{23} \right].$$

A BSM on 2, 3 projects 1, 4 onto the *same* Bell state — **photons 1 and 4 become entangled although they never interacted**. Swapping is teleportation of an entangled half: photon 2's entanglement with 1 is teleported onto photon 4. Since the outcome is announced, Bob can rotate to a standard Bell state.

💡 Tip

Entanglement swapping is the core of **quantum repeaters**. Direct transmission decays exponentially with fibre length; dividing the distance into segments, entangling neighbouring nodes and swapping at intermediate stations (with entanglement purification and quantum memories) makes the scaling

polynomial. It is also used in measurement-device-independent QKD, where an untrusted middle station performs the BSM.

11.5. Exercises

Exercise 11.1

Verify the teleportation identity for the $|\Psi^-\rangle_{CA}$ term and find the correction Bob must apply.

Solution

Collect in $|\psi\rangle_C |\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle)_{CAB}$ the terms with $CA \in \{01, 10\}$: $\frac{1}{\sqrt{2}}(\alpha|01\rangle|1\rangle + \beta|10\rangle|0\rangle)$. Using $|01\rangle = (|\Psi^+\rangle + |\Psi^-\rangle)/\sqrt{2}$ and $|10\rangle = (|\Psi^+\rangle - |\Psi^-\rangle)/\sqrt{2}$, the $|\Psi^-\rangle$ component is $\frac{1}{2}|\Psi^-\rangle(\alpha|1\rangle - \beta|0\rangle)$. Applying $\hat{X}$ gives $\alpha|0\rangle - \beta|1\rangle$, then $\hat{Z}$ gives $\alpha|0\rangle + \beta|1\rangle$: the correction is $\hat{Z}\hat{X}$.

Exercise 11.2

Show that, averaged over Alice's outcomes, Bob's state before the classical message is $\hat{\mathbb{1}}/2$.

Solution

Bob's state is $\frac{1}{4} \sum_{k=0}^3 \hat{\sigma}_k |\psi\rangle \langle \psi| \hat{\sigma}_k$ with $\hat{\sigma}_k \in \{\hat{\mathbb{1}}, \hat{Z}, \hat{X}, \hat{X}\hat{Z}\}$. In Bloch terms the four Paulis flip pairs of components of $\vec{r}$, and the average of $\vec{r}$ over the four is 0; equivalently, this is the completely depolarizing channel: $\frac{1}{4}(\hat{\rho} + \hat{X}\hat{\rho}\hat{X} + \hat{Y}\hat{\rho}\hat{Y} + \hat{Z}\hat{\rho}\hat{Z}) = \hat{\mathbb{1}}/2$.

11.6. Sources

Prof. Vallone, *Quantum Information and Computing*, lecture 15 (Bell-state measurement with beam splitter and PBS, teleportation, entanglement swapping) and the course outline (dense coding). Nielsen & Chuang, Sections 1.3.7 and 2.3. Benenti, Casati & Strini, Vol. I, Chapter 4. Original references: C. Bennett *et al.*, Phys. Rev. Lett. **70**, 1895 (1993); C. Bennett and S. Wiesner, Phys. Rev. Lett. **69**, 2881 (1992); M. Żukowski *et al.*, Phys. Rev. Lett. **71**, 4287 (1993); D. Bouwmeester *et al.*, Nature **390**, 575 (1997).

Part III.

Part III: Quantum computation

12. Quantum circuits and universality

To talk about quantum *computation* we need a model of computation. The standard one is the **circuit model**: a register of qubits, a sequence of elementary gates, a final measurement. This chapter recalls the classical background — universal gate sets, complexity classes, reversible computation — then introduces quantum gates, proves that a small set of them is universal, and shows the first glimpse of quantum speed-up through quantum parallelism and the Deutsch–Jozsa algorithm.

12.1. Classical circuits and complexity

A classical computation evaluates a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ with a circuit of elementary gates (AND, OR, NOT, XOR, FANOUT). A gate set is **universal** if every Boolean function can be built from it: $\{\text{AND}, \text{OR}, \text{NOT}\}$ is universal (write f in disjunctive normal form), and so is NAND alone (with FANOUT).

The cost of a computation is measured by how the number of gates (time) or memory (space) scales with the input size n :

- **P**: problems solvable by a deterministic algorithm in time polynomial in n — “efficiently solvable”;
- **NP**: problems whose solutions can be *verified* in polynomial time (factoring, satisfiability, travelling salesman decision version); whether $P = NP$ is open;
- **BPP**: solvable in polynomial time by a randomised algorithm with error probability $\leq \frac{1}{3}$;
- **PSPACE**: solvable with polynomial memory;
- **BQP**: solvable in polynomial time by a quantum computer with bounded error.

Known inclusions: $P \subseteq BPP \subseteq BQP \subseteq PSPACE$. Factoring is in NP (and coNP) and in BQP (Shor, Chapter 13), but no classical polynomial algorithm is known; it is *not* believed to be NP-complete. Quantum computers are not expected to solve NP-complete problems efficiently.

12.2. Reversible computation

Quantum evolution is unitary, hence reversible; classical gates such as AND are not (two input bits, one output bit: information is erased). **Landauer’s principle** states that erasing one bit dissipates at least $k_B T \ln 2$ of heat, so irreversibility has a physical cost.

Definition 12.1 (Toffoli gate). The **Toffoli** (controlled-controlled-NOT) gate maps $(a, b, c) \mapsto (a, b, c \oplus ab)$. It is reversible (it is its own inverse) and universal for classical computation: with $c = 1$ it computes NAND, with $a = 1, c = 0$ it copies b (FANOUT).

Any irreversible circuit can be made reversible with ancilla bits: $(x, 0, 0) \mapsto (x, f(x), g(x))$, where $g(x)$ is “garbage”. Bennett’s trick removes the garbage: compute f , copy the result with CNOTs onto a fresh register, and run the computation backwards. The resulting map

$$(x, y) \mapsto (x, y \oplus f(x))$$

is the canonical reversible form of any function, with only polynomial overhead. Its quantum version is the oracle $\hat{U}_f$ below.

12.3. The quantum circuit model

A quantum computation acts on n qubits, initialised in $|0\rangle^{\otimes n}$, applies a sequence of gates (unitaries acting on one or two qubits at a time), and ends with measurements in the computational basis. Common gates:

gate	matrix / action
Pauli X, Y, Z	bit flip, $Y = iXZ$, phase flip
Hadamard H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
phase S	$\text{diag}(1, i)$
T ($\pi/8$)	$\text{diag}(1, e^{i\pi/4})$
rotation $R_{\vec{n}}(\theta)$	$e^{-i\theta \vec{n} \cdot \hat{\sigma}}/2$
CNOT	$ a, b\rangle \mapsto a, b \oplus a\rangle$
controlled- U	$ 0\rangle\langle 0 \otimes \hat{1} + 1\rangle\langle 1 \otimes \hat{U}$
Toffoli	$ a, b, c\rangle \mapsto a, b, c \oplus ab\rangle$

The CNOT creates entanglement from product states. The basic example prepares Bell states:

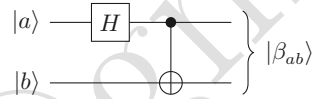


Figure: Hadamard followed by CNOT maps $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ to $|\Phi^+\rangle, |\Psi^+\rangle, |\Phi^-\rangle, |\Psi^-\rangle$; run backwards, it is the Bell-state measurement of Chapter 11.

12.4. Universality

Theorem 12.1 (Exact universality). *Single-qubit gates together with CNOT are **universal**: any unitary on n qubits can be written exactly as a product of them.*

Proof sketch. (1) Any $d \times d$ unitary is a product of at most $d(d-1)/2$ **two-level unitaries**, each acting non-trivially only on two basis states (a Givens-rotation argument, like Gaussian elimination). (2) A two-level unitary on basis states $|s\rangle, |t\rangle$ is implemented by permuting the basis with CNOT/Toffoli gates along a Gray code from s to t , applying a multi-controlled single-qubit gate, and undoing the permutation. (3) Multi-controlled gates reduce to Toffolis and controlled- U gates; controlled- U reduces to CNOTs and single-qubit gates via the decomposition $U = e^{i\alpha}AXBXC$ with $ABC = \hat{1}$. $\square$

The construction can need exponentially many gates: universality says every unitary is *possible*, not that it is *efficient*. Indeed most unitaries require exponentially many gates; quantum algorithms are the rare ones that do not.

Real hardware offers a finite set of gates. A continuum of unitaries cannot be reached exactly, but can be approximated.

Theorem 12.2 (Approximate universality and Solovay–Kitaev). *The finite set $\{H, T, CNOT\}$ (with $S = T^2$) is universal in the approximate sense: any unitary can be approximated to accuracy ε . Moreover, by the Solovay–Kitaev theorem, a single-qubit gate needs only $O(\log^c(1/\varepsilon))$ gates from the set, with $c \approx 2$: approximation is cheap.*

H and T generate rotations about two non-parallel axes by angles that are irrational multiples of π , which fill the Bloch sphere densely. A circuit of m gates approximated to ε/m each has total error at most ε (errors add up linearly for unitaries).

12.5. Quantum parallelism and the first algorithm

Given a classical function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, define the reversible **oracle**

$$\hat{U}_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle.$$

Applied to a uniform superposition — prepared with n Hadamards, $H^{\otimes n} |0\rangle^{\otimes n} = 2^{-n/2} \sum_x |x\rangle$ — it gives

$$\hat{U}_f \frac{1}{2^{n/2}} \sum_x |x\rangle |0\rangle = \frac{1}{2^{n/2}} \sum_x |x\rangle |f(x)\rangle :$$

all 2^n values of f in a single evaluation. This **quantum parallelism** is not directly useful — measuring returns one random pair $(x, f(x))$ — but interference can extract *global* properties of f .

Phase kickback. With the target in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$,

$$\hat{U}_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle,$$

so the function value appears as a phase on the control register.

Example 12.1 (Deutsch–Jozsa algorithm)

Promise: $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is either **constant** or **balanced** (equal to 1 on exactly half the inputs). Classically, a deterministic algorithm may need $2^{n-1} + 1$ evaluations. Quantum:

1. start in $|0\rangle^{\otimes n} |1\rangle$ and apply $H^{\otimes(n+1)}: 2^{-n/2} \sum_x |x\rangle |-\rangle$;
2. apply $\hat{U}_f: 2^{-n/2} \sum_x (-1)^{f(x)} |x\rangle |-\rangle$;
3. apply $H^{\otimes n}$ to the first register, using $H^{\otimes n} |x\rangle = 2^{-n/2} \sum_z (-1)^{x \cdot z} |z\rangle$; the amplitude of $|0\rangle^{\otimes n}$ is

$$\frac{1}{2^n} \sum_x (-1)^{f(x)} = \begin{cases} \pm 1 & f \text{ constant,} \\ 0 & f \text{ balanced;} \end{cases}$$

4. measure: all zeros iff f is constant.

One query decides with certainty. For $n = 1$ this is Deutsch’s algorithm (is $f(0) = f(1)$?). The separation disappears against randomised classical algorithms with small error, but the structure — superposition, oracle with phase kickback, interference by a Fourier-type transform — is the template of Shor’s algorithm.

12.6. Exercises

Exercise 12.1

Show that $\text{CNOT} = (\hat{1} \otimes H) \text{CZ} (\hat{1} \otimes H)$, where $\text{CZ} = \text{diag}(1, 1, 1, -1)$, and that CZ is symmetric in its two qubits.

Solution

CZ applies Z to the target when the control is 1. Since $HZH = X$, conjugating the target by H turns controlled- Z into controlled- X . CZ is diagonal with a -1 only on $|11\rangle$, which is symmetric under exchanging the qubits: “control” and “target” are interchangeable.

Exercise 12.2

Construct a circuit of CNOTs that swaps two qubits.

Solution

Three alternating CNOTs: $\text{SWAP} = \text{CNOT}_{12} \text{CNOT}_{21} \text{CNOT}_{12}$. On basis states: $(a, b) \rightarrow (a, a \oplus b) \rightarrow (b, a \oplus b) \rightarrow (b, a)$, the XOR-swap trick.

12.7. Sources

Prof. Vallone, *Quantum Information and Computing*, lectures 16 (universal classical gates, complexity classes, reversibility, circuit model) and 17 (universal gates for quantum computation), and lecture 19 (function evaluation). Nielsen & Chuang, Sections 1.3–1.4, 3.2, 4.1–4.5 and Appendix 3 (Solovay–Kitaev). Benenti, Casati & Strini, Vol. I, Chapters 1 and 3.

13. Fourier transform, phase estimation and Shor's algorithm

The most celebrated quantum algorithm factors integers in polynomial time, breaking RSA. Its engine is the **quantum Fourier transform** (QFT), which turns phases into measurable bit strings, packaged in the **phase estimation** algorithm. Factoring reduces classically to finding the period of modular exponentiation, and phase estimation finds periods. This chapter builds the three layers: QFT, phase estimation, order finding and Shor's algorithm.

13.1. The quantum Fourier transform

Definition 13.1 (Quantum Fourier transform). On an n -qubit register with $N = 2^n$ basis states $|j\rangle$, $j = 0, \dots, N - 1$,

$$\text{QFT } |j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j k / N} |k\rangle.$$

It is the discrete Fourier transform applied to the amplitudes, and it is unitary. For $n = 1$ it is the Hadamard gate. Writing $j = j_1 j_2 \dots j_n$ in binary and $0.j_\ell j_{\ell+1} \dots j_n = \sum_m j_m 2^{\ell-1-m}$ for binary fractions, the QFT factorises:

$$\text{QFT } |j_1 \dots j_n\rangle = \frac{1}{2^{n/2}} (|0\rangle + e^{2\pi i 0.j_n} |1\rangle) \otimes (|0\rangle + e^{2\pi i 0.j_{n-1} j_n} |1\rangle) \otimes \dots \otimes (|0\rangle + e^{2\pi i 0.j_1 j_2 \dots j_n} |1\rangle).$$

Each output qubit is a product state whose phase depends on some of the input bits. This gives the circuit: on qubit j_1 apply H (producing the phase $e^{2\pi i 0.j_1}$), then controlled phase gates $R_k = \text{diag}(1, e^{2\pi i / 2^k})$ controlled by $j_2, \dots, j_n$ (adding $0.0j_2, 0.00j_3, \dots$); repeat on $j_2, j_3, \dots$; finally reverse the qubit order with swaps. The total is $n(n+1)/2$ gates, i.e. $O(n^2)$, against $O(n2^n)$ for the classical FFT on 2^n numbers.

Warning

The QFT does not make classical Fourier analysis exponentially faster: the transformed amplitudes cannot be read out (a measurement returns one k at random). It is useful only when the output is concentrated on a few values carrying the information we need — exactly the situation in phase estimation.

13.2. Phase estimation

Problem. A unitary $\hat{U}$ has an eigenvector $|u\rangle$ with eigenvalue $e^{2\pi i \varphi}$, $\varphi \in [0, 1)$ unknown. Given the ability to prepare $|u\rangle$ and to apply controlled- $\hat{U}^{2^j}$ gates, estimate φ to t bits.

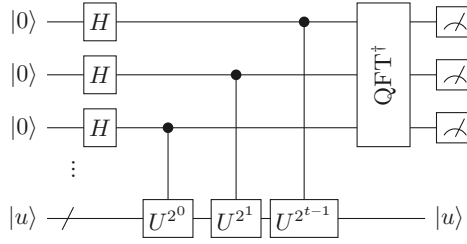


Figure: phase estimation with a t -qubit control register (three wires shown) and a target register prepared in the eigenvector $|u\rangle$.

Analysis. After the Hadamards, the control register is $2^{-t/2} \sum_k |k\rangle$. A controlled- $\hat{U}^{2^j}$ acting on $|u\rangle$ kicks back the phase $e^{2\pi i \varphi 2^j}$ onto control qubit j (phase kickback, as in Chapter 12), so the state becomes

$$\frac{1}{2^{t/2}} \sum_{k=0}^{2^t-1} e^{2\pi i \varphi k} |k\rangle \otimes |u\rangle.$$

This is exactly $\text{QFT} |2^t \varphi\rangle$. If $\varphi = 0.\varphi_1 \dots \varphi_t$ has an exact t -bit expansion, the inverse QFT returns $|\varphi_1 \dots \varphi_t\rangle$ with certainty. Otherwise the measured value is the best t -bit approximation with probability at least $4/\pi^2 \approx 0.405$; with

$$t = n + \left\lceil \log_2 \left(2 + \frac{1}{2\varepsilon} \right) \right\rceil$$

control qubits one obtains φ to n bits with probability at least $1 - \varepsilon$.

If the target is a superposition $\sum_u c_u |u\rangle$ of eigenvectors, the algorithm returns the phase φ_u with probability $|c_u|^2$ — a key point for order finding.

13.3. Order finding

Definition 13.2 (Order). For integers $x < N$ with $\gcd(x, N) = 1$, the **order** of x modulo N is the smallest $r > 0$ such that $x^r \equiv 1 \pmod{N}$.

No efficient classical algorithm is known. Quantumly, consider the unitary $\hat{U}_x |y\rangle = |xy \bmod N\rangle$ on $L = \lceil \log_2 N \rceil$ qubits (a permutation of basis states with $y < N$). Its eigenvectors include

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |x^k \bmod N\rangle, \quad \hat{U}_x |u_s\rangle = e^{2\pi i s / r} |u_s\rangle, \quad s = 0, \dots, r-1.$$

Phase estimation on $|u_s\rangle$ would yield s/r . We cannot prepare a specific $|u_s\rangle$ without knowing r — but we do not need to, since

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle.$$

Running phase estimation on $|1\rangle$ returns an estimate of s/r for a uniformly random s . The **continued fractions algorithm** extracts the fraction s/r from a $2L + 1$ -bit approximation; if $\gcd(s, r) = 1$ (which happens with probability $\Omega(1/\log \log r)$, so a few repetitions suffice) the denominator is r .

The controlled- $\hat{U}_x^{2^j}$ gates are implemented by **modular exponentiation**: compute $x^{2^j} \bmod N$ by repeated squaring, then multiply. Its cost, $O(L^3)$ gates with schoolbook multiplication, dominates the algorithm.

13.4. Shor's algorithm

Theorem 13.1 (Reduction of factoring to order finding). *Let N be odd and not a prime power, and x chosen uniformly among the integers in $[2, N-1]$ coprime to N . With probability at least $\frac{1}{2}$, the order r of x is even and $x^{r/2} \not\equiv -1 \pmod{N}$. In that case*

$$\gcd(x^{r/2} - 1, N) \quad \text{and} \quad \gcd(x^{r/2} + 1, N)$$

are non-trivial factors of N .

Reason. $x^r - 1 = (x^{r/2} - 1)(x^{r/2} + 1) \equiv 0 \pmod{N}$, so N divides the product. It does not divide $x^{r/2} - 1$ (by minimality of r) nor, by assumption, $x^{r/2} + 1$; hence it shares a non-trivial factor with each. $\square$

Shor's algorithm for an L -bit N :

1. If N is even, return 2. If $N = a^b$, return a (classically efficient check).
2. Pick a random $x \in [2, N-1]$. If $\gcd(x, N) > 1$, return it (Euclid's algorithm).
3. Find the order r of x with the quantum order-finding subroutine.
4. If r is odd or $x^{r/2} \equiv -1 \pmod{N}$, go back to 2. Otherwise return $\gcd(x^{r/2} \pm 1, N)$.

The quantum part costs $O(L^3)$ operations (less with fast multiplication); the best known classical algorithm, the number field sieve, runs in time $\exp(O(L^{1/3}(\log L)^{2/3}))$ — a super-polynomial speed-up. RSA-2048 would require millions of noisy physical qubits with error correction; current devices are far from that, which is why post-quantum cryptography and QKD (Chapter 10) are being deployed now.

Example 13.1 (Factoring 15)

Take $x = 7$: the powers modulo 15 are 7, 4, 13, 1, so $r = 4$. Phase estimation returns $s/4 \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ with equal probability; the outcomes $\frac{1}{4}$ and $\frac{3}{4}$ give $r = 4$ directly. Then $x^{r/2} = 49 \equiv 4 \pmod{15}$, $\gcd(3, 15) = 3$ and $\gcd(5, 15) = 5$. This instance was run on an NMR quantum computer (2001) and on a scalable trapped-ion processor (2016).

13.5. Exercises

Exercise 13.1

Write the QFT on two qubits as a 4×4 matrix and check that it maps $|00\rangle$ to the uniform superposition.

Solution

With $\omega = e^{2\pi i/4} = i$: $\text{QFT}_4 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$. The first column gives $\text{QFT}|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

Exercise 13.2

Factor $N = 21$ with $x = 2$. What could go wrong with $x = 20$?

Solution

Powers of 2 mod 21: 2, 4, 8, 16, 11, 1, so $r = 6$. $2^3 = 8 \not\equiv -1$; $\gcd(7, 21) = 7$ and $\gcd(9, 21) = 3$. With $x = 20 \equiv -1$: $r = 2$ and $x^{r/2} = 20 \equiv -1 \pmod{21}$, the bad case — the algorithm must restart with

another x .

13.6. Sources

Prof. Vallone, *Quantum Information and Computing*, lectures 18 (quantum Fourier transform, Shor's algorithm) and 19 (phase estimation). Nielsen & Chuang, Chapter 5 and Appendix 4. Benenti, Casati & Strini, Vol. I, Chapter 3. Further reading suggested in the lectures: Phys. Rev. Lett. **85**, 3049 (2000); T. Monz *et al.*, *Realization of a scalable Shor algorithm*, Science **351**, 1068 (2016). Original reference: P. Shor, *Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer*, SIAM J. Comput. **26**, 1484 (1997).

14. Grover's search algorithm

Shor's algorithm exploits a hidden periodic structure. Most problems have no such structure: the generic task is to find a marked item among N possibilities, which classically requires about N checks. Grover's algorithm does it with $O(\sqrt{N})$ queries — a quadratic speed-up that is provably optimal and applies to any problem whose solutions can be recognised efficiently. Its analysis reduces to a rotation in a two-dimensional plane.

14.1. The search problem and the oracle

Let $N = 2^n$ items be labelled by $x \in \{0, \dots, N-1\}$, and $f(x) = 1$ for the M solutions, 0 otherwise. We can evaluate f but know nothing else about it. A classical algorithm needs $O(N/M)$ evaluations.

Quantum access to f is the oracle $\hat{U}_f |x\rangle |q\rangle = |x\rangle |q \oplus f(x)\rangle$. With the oracle qubit in $|-\rangle$, phase kickback (Chapter 12) gives the **phase oracle**

$$\hat{O}|x\rangle = (-1)^{f(x)}|x\rangle,$$

which flips the sign of the solutions. Recognising a solution must be efficient — e.g. checking a candidate assignment of a SAT formula — while finding one may be hard.

14.2. The Grover iteration

Start from the uniform superposition $|\psi\rangle = H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_x |x\rangle$ and repeat the **Grover iteration**

$$\hat{G} = (2|\psi\rangle\langle\psi| - \hat{1})\hat{O}.$$

The **diffusion** operator $2|\psi\rangle\langle\psi| - \hat{1} = H^{\otimes n}(2|0\rangle\langle 0| - \hat{1})H^{\otimes n}$ is an “inversion about the mean”: it maps $\sum_x a_x |x\rangle$ to $\sum_x (2\bar{a} - a_x) |x\rangle$ with $\bar{a}$ the average amplitude. After the oracle has made the solutions' amplitudes negative, the inversion about the mean amplifies them.

14.3. Geometric picture

Define the normalised superpositions of non-solutions and solutions,

$$|\alpha\rangle = \frac{1}{\sqrt{N-M}} \sum_{x: f(x)=0} |x\rangle, \quad |\beta\rangle = \frac{1}{\sqrt{M}} \sum_{x: f(x)=1} |x\rangle.$$

The initial state lies in their plane:

$$|\psi\rangle = \cos \frac{\theta}{2} |\alpha\rangle + \sin \frac{\theta}{2} |\beta\rangle, \quad \sin \frac{\theta}{2} = \sqrt{\frac{M}{N}}.$$

14. Grover's search algorithm

In this plane, $\hat{O}$ is a reflection about $|\alpha\rangle$ and $2|\psi\rangle\langle\psi| - \hat{1}$ a reflection about $|\psi\rangle$. Two reflections compose to a **rotation by twice the angle between the axes**, i.e. by θ :

$$\hat{G}^k |\psi\rangle = \cos\left(\frac{2k+1}{2}\theta\right) |\alpha\rangle + \sin\left(\frac{2k+1}{2}\theta\right) |\beta\rangle.$$

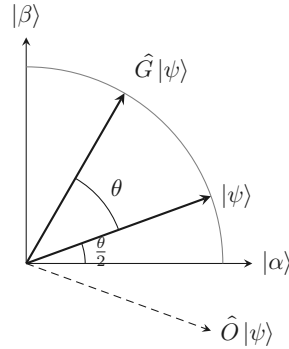


Figure: the oracle reflects $|\psi\rangle$ about $|\alpha\rangle$; the diffusion reflects it back about $|\psi\rangle$; the net effect is a rotation by θ towards the solutions.

14.4. Number of iterations and success probability

We want $\frac{2k+1}{2}\theta \approx \frac{\pi}{2}$, i.e. to rotate $|\psi\rangle$ by the angle $\arccos\sqrt{M/N}$ that separates it from $|\beta\rangle$. The best number of iterations is the closest integer

$$R = \text{CI}\left(\frac{\arccos\sqrt{M/N}}{\theta}\right) \approx \frac{\pi}{4}\sqrt{\frac{N}{M}} \quad (M \ll N).$$

After R iterations the state is within angle $\theta/2$ of $|\beta\rangle$, so the probability of measuring a solution is at least $1 - M/N$. Each iteration uses one oracle call: $O(\sqrt{N/M})$ **queries** against $O(N/M)$ classically.

Overshooting is harmful: the rotation continues past $|\beta\rangle$ and the success probability oscillates. If M is unknown, one can estimate it first with **quantum counting** (phase estimation applied to $\hat{G}$, whose eigenvalues are $e^{\pm i\theta}$) or use randomised iteration counts.

Example 14.1 (Four items, one solution)

$N = 4$, $M = 1$: $\sin\frac{\theta}{2} = \frac{1}{2}$, so $\theta = \frac{\pi}{3}$. After one iteration the angle is $\frac{3}{2}\theta = \frac{\pi}{2}$: the state is $|\beta\rangle$ and the solution is found with certainty after a single oracle query. In amplitudes: start with all $\frac{1}{2}$; the oracle gives $(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$ (solution third), mean $\frac{1}{4}$; inversion about the mean gives $(0, 0, 1, 0)$. Classically one needs on average 2.25 queries.

Theorem 14.1 (Optimality (Bennett, Bernstein, Brassard, Vazirani, 1997)). *Any quantum algorithm that finds a marked item among N with bounded error using an oracle needs $\Omega(\sqrt{N})$ oracle queries. Grover's algorithm is optimal.*

Hence quantum computers cannot solve NP-complete problems by brute-force search in polynomial time: a search over 2^n assignments still takes $2^{n/2}$ steps. The quadratic speed-up is nonetheless broad: it applies to any NP problem, to minimum finding, and — as **amplitude amplification** — boosts the success probability p of any quantum subroutine with $O(1/\sqrt{p})$ repetitions instead of $O(1/p)$. In cryptography it halves the effective key length of symmetric ciphers (AES-128 offers ≈ 64 -bit security against Grover), which is why AES-256 is recommended in the post-quantum era.

14.5. Exercises

Exercise 14.1

For $N = 2^{10}$ and $M = 1$, how many Grover iterations are optimal and what is the success probability?

Solution

$\sin \frac{\theta}{2} = 2^{-5}$, so $\theta \approx 0.0625$ rad. $R = \text{CI}(\arccos(2^{-5})/\theta) = \text{CI}(1.5395/0.0625) = 25$. After 25 iterations the angle is $\frac{51}{2}\theta \approx 1.594$ rad, so $p = \sin^2(1.594) \approx 0.9995$. Classically about 512 queries are needed on average.

Exercise 14.2

Show that $2|\psi\rangle\langle\psi| - \hat{1}$ maps amplitudes a_x to $2\bar{a} - a_x$.

Solution

$\langle\psi|\phi\rangle = \frac{1}{\sqrt{N}} \sum_x a_x = \sqrt{N} \bar{a}$, so $2|\psi\rangle\langle\psi|\phi\rangle = 2\sqrt{N}\bar{a} \cdot \frac{1}{\sqrt{N}} \sum_x |x\rangle = \sum_x 2\bar{a}|x\rangle$. Subtracting $|\phi\rangle$ gives $\sum_x (2\bar{a} - a_x)|x\rangle$.

14.6. Sources

Prof. Vallone, *Quantum Information and Computing*, lecture 19 (Grover algorithm). Nielsen & Chuang, Chapter 6. Benenti, Casati & Strini, Vol. I, Chapter 3. Original references: L. Grover, *A fast quantum mechanical algorithm for database search*, Proc. STOC (1996), and Phys. Rev. Lett. **79**, 325 (1997); C. Bennett, E. Bernstein, G. Brassard and U. Vazirani, SIAM J. Comput. **26**, 1510 (1997).

15. Error correction and physical implementations

Quantum algorithms assume perfect gates on perfectly isolated qubits. Real qubits decohere (Chapter 8) and real gates are imprecise. Without a way to fight errors, quantum computation would be a theoretical curiosity. This chapter shows that errors *can* be corrected despite no-cloning and measurement disturbance, sketches the threshold theorem, surveys alternative computing paradigms (one-way, adiabatic, quantum simulation, boson sampling), and reviews the main physical platforms.

15.1. Why quantum error correction seems impossible

Classical memories protect bits by redundancy: encode $0 \rightarrow 000$, $1 \rightarrow 111$ and decode by majority vote; if each bit flips independently with probability p , the logical error rate drops to $3p^2 - 2p^3$. Three obstacles seem to forbid a quantum analogue:

1. **No-cloning:** we cannot encode $|\psi\rangle \rightarrow |\psi\rangle |\psi\rangle |\psi\rangle$.
2. **Continuous errors:** errors are not just flips but arbitrary small rotations and decoherence.
3. **Measurement destroys:** reading the qubits to diagnose an error collapses the superposition.

All three are circumvented.

15.2. The three-qubit bit-flip code

Encode by entangling, not copying:

$$\alpha|0\rangle + \beta|1\rangle \mapsto \alpha|000\rangle + \beta|111\rangle = \alpha|0_L\rangle + \beta|1_L\rangle.$$

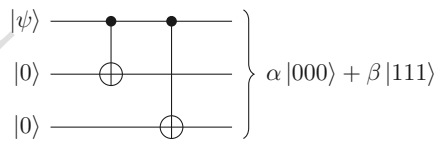


Figure: encoding circuit of the bit-flip code — two CNOTs, no copying.

Suppose at most one qubit suffers an X error. Measure the two **parity checks** Z_1Z_2 and Z_2Z_3 (with ancillas and CNOTs, without measuring individual qubits):

syndrome (Z_1Z_2, Z_2Z_3)	diagnosis	correction
(+1, +1)	no error	none
(-1, +1)	qubit 1 flipped	X_1
(-1, -1)	qubit 2 flipped	X_2
(+1, -1)	qubit 3 flipped	X_3

The parities have the *same* value on $|000\rangle$ and $|111\rangle$, so the **syndrome measurement reveals the error but nothing about α, β** : the superposition survives. The logical error rate is $O(p^2)$.

Continuous errors are discretised. A small rotation $e^{-i\epsilon X_1} = \cos \epsilon \hat{1} - i \sin \epsilon X_1$ produces a superposition of “no error” and “ X_1 error”. The syndrome measurement collapses it onto one of the two, and the correction fixes either. More generally:

Theorem 15.1 (Discretisation of errors). *If a code corrects a set of error operators $\{\hat{E}_k\}$, it corrects every error that is a linear combination of them — in particular every quantum channel whose Kraus operators lie in their span. Correcting X , $Y = iXZ$, Z on each qubit (and $\hat{1}$) suffices to correct arbitrary single-qubit errors.*

15.3. Phase flips and the Shor code

A phase flip Z is a bit flip in the $\{|\pm\rangle\}$ basis, so the **phase-flip code** $|0_L\rangle = |+++ \rangle$, $|1_L\rangle = |--\rangle$ (encode as before, then apply $H^{\otimes 3}$) corrects one Z error with syndromes X_1X_2 , X_2X_3 .

Shor’s nine-qubit code (1995) concatenates the two:

$$|0_L\rangle = \frac{(|000\rangle + |111\rangle)^{\otimes 3}}{2\sqrt{2}}, \quad |1_L\rangle = \frac{(|000\rangle - |111\rangle)^{\otimes 3}}{2\sqrt{2}}.$$

Each block of three corrects a bit flip; the relative signs between blocks correct a phase flip. It corrects an arbitrary error on any single qubit — the first proof that quantum error correction is possible. Smaller codes exist (Steane’s 7-qubit code, the 5-qubit perfect code), and all of them fit the **stabilizer formalism**: the code space is the joint +1 eigenspace of a group of commuting Pauli products, whose measured eigenvalues form the syndrome.

Theorem 15.2 (Threshold theorem). *If every physical component (gate, preparation, measurement, idle step) fails independently with probability below a threshold p_{th} , then arbitrarily long quantum computations can be performed reliably, with overhead polylogarithmic in the size of the computation, using concatenated fault-tolerant codes.*

Thresholds range from $\sim 10^{-4}$ for concatenated codes to $\sim 1\%$ for the **surface code**, a two-dimensional lattice code with only nearest-neighbour parity checks — the current front-runner, whose below-threshold operation has been demonstrated on superconducting processors. The overhead is still large: thousands of physical qubits per logical qubit for algorithms like Shor’s.

15.4. Other paradigms of quantum computation

One-way (measurement-based) computing. Prepare a highly entangled **cluster state** (qubits in $|+\rangle$ entangled by CZ gates on a lattice), then compute by single-qubit measurements only, choosing each measurement basis from earlier outcomes (feed-forward). Teleportation-like corrections propagate the logical state; the entanglement of the cluster is the resource consumed. It is natural for photons, where entangling gates are probabilistic but measurements are easy.

Adiabatic quantum computing. Encode the answer in the ground state of a problem Hamiltonian $\hat{H}_P$; start in the easy ground state of $\hat{H}_0$ and slowly interpolate $\hat{H}(t) = (1 - t/T)\hat{H}_0 + (t/T)\hat{H}_P$. By the adiabatic theorem the system stays in the ground state if $T = O(1/g^2)$, where g is the minimum spectral gap along the path. It is polynomially equivalent to the circuit model; the hard instances are those with exponentially small gaps. **Quantum annealing** (D-Wave) is its noisy, finite-temperature heuristic version for optimisation problems, using quantum fluctuations to escape local minima — sometimes outperforming simulated annealing, with no proven general advantage.

Quantum simulation. Feynman’s original motivation: use a controllable quantum system to simulate another. *Digital* simulation decomposes $e^{-i\hat{H}t}$ into gates (Trotter formulas); *analog* simulation engineers a device whose Hamiltonian mimics the target. Ultracold atoms in optical lattices realise the Bose–Hubbard model

$$\hat{H} = -J \sum_{\langle i,j \rangle} \hat{a}_i^\dagger \hat{a}_j + \frac{U}{2} \sum_i \hat{n}_i(\hat{n}_i - 1),$$

(J hopping between neighbouring sites, U on-site interaction), observing the superfluid–Mott-insulator transition.

Boson sampling and quantum advantage. Sending n indistinguishable photons through a random m -mode linear interferometer and sampling the output pattern is believed classically hard (the amplitudes are matrix permanents, which are #P-hard; Aaronson–Arkhipov 2011), yet it needs no entangling gates. Boson sampling and random-circuit sampling (Google’s *Sycamore*, 2019) were the first experiments claiming **quantum computational advantage** — on tasks of no practical use, and with classical simulation algorithms continually narrowing the gap. Reprogrammable linear-optical circuits can implement any linear-optical protocol up to their size.

15.5. Physical implementations

DiVincenzo’s criteria for a quantum computer: (1) scalable, well-characterised qubits; (2) initialisation to a simple state; (3) coherence times much longer than gate times; (4) a universal gate set; (5) qubit-specific measurement (plus, for networks, interconversion with flying qubits and their faithful transmission). Main platforms:

- **Photons:** long coherence, room temperature, natural for communication; two-photon gates are probabilistic (KLM scheme with linear optics, ancillas and feed-forward, 2001) — hence measurement-based and fusion-based architectures.
- **Trapped ions:** qubits in hyperfine or optical levels, gates via laser-driven collective motion; record fidelities and all-to-all connectivity; slower gates, scaling via shuttling or photonic links.
- **Neutral atoms** in optical lattices or tweezer arrays: large, reconfigurable arrays, Rydberg-mediated gates, natural for analog simulation.
- **NMR:** nuclear spins in molecules, manipulated by RF pulses; the first demonstrations of quantum algorithms (Shor’s factoring of 15 in 2001) but not scalable (ensemble, pseudo-pure states).
- **Quantum dots:** electron spins in semiconductors, compatible with silicon fabrication.
- **Superconducting qubits:** Josephson-junction circuits (transmons) at millikelvin, microwave control and readout, fast gates; the basis of IBM’s cloud-accessible processors (programmable through the IBM Quantum Composer) and of Google’s devices.
- **Topological qubits** (pursued by Microsoft): information stored non-locally in exotic quasiparticles, intrinsically protected from local noise — still at an early experimental stage.

Today’s processors belong to the **NISQ** era (noisy intermediate-scale quantum): tens to hundreds of qubits without full error correction, used for variational algorithms, simulation and benchmarking while fault tolerance is developed.

15.6. Exercises

Exercise 15.1

For the bit-flip code with independent flip probability p per qubit and perfect correction of single flips, show that the logical error probability is $3p^2 - 2p^3$ and find when encoding helps.

Solution

The code fails iff two or three qubits flip: $3p^2(1 - p) + p^3 = 3p^2 - 2p^3$. This is smaller than p iff $3p - 2p^2 < 1$, i.e. $p < \frac{1}{2}$.

Exercise 15.2

Show that the syndrome measurement of $Z_1 Z_2$ does not disturb $\alpha|000\rangle + \beta|111\rangle$, while measuring Z_1 would.

Solution

Both $|000\rangle$ and $|111\rangle$ are $+1$ eigenstates of $Z_1 Z_2$, so the encoded state is an eigenstate and the measurement leaves it unchanged. Z_1 has eigenvalue $+1$ on $|000\rangle$ and -1 on $|111\rangle$: measuring it collapses the state to $|000\rangle$ or $|111\rangle$, destroying the superposition.

15.7. Sources

Prof. Vallone, *Quantum Information and Computing*, lecture 21 (quantum error correction; one-way, adiabatic, quantum simulation, boson sampling; physical implementations) and the deck **QComputing_25.26** (quantum simulation, adiabatic computing, one-way computing, photons, trapped ions, cold atoms, NMR, quantum dots, superconducting qubits, IBM, Microsoft, D-Wave, quantum supremacy). Nielsen & Chuang, Chapters 7 and 10. Benenti, Casati & Strini, Vol. I, Chapters 7–8. Further reading: T. Ladd *et al.*, *Quantum computers*, Nature **464**, 45 (2010); K. Bharti *et al.*, *Noisy intermediate-scale quantum algorithms*, Rev. Mod. Phys. **94**, 015004 (2022); E. Knill, R. Laflamme and G. Milburn, Nature **409**, 46 (2001); J. Carolan *et al.*, *Universal linear optics*, Science **349**, 711 (2015); F. Arute *et al.*, Nature **574**, 505 (2019).